

1 Embedded Artificial Intelligence in Battery Management
2 Systems: Pruning and Quantization for Efficient
3 State-of-Charge and State-of-Health Estimation

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6 **Abstract**

7 Online battery management requires accurate state-of-charge (SOC) and state-
8 of-health (SOH) estimates within strict memory and timing budgets. To address
9 these constraints, this work investigates how stateful long short-term memory
10 (LSTM) networks with multi-layer perceptron (MLP) heads can be optimised
11 for embedded artificial intelligence (AI), using structured pruning and recurrent-
12 weight quantization to reduce memory and computational costs while retain-
13 ing estimation accuracy on an STM32H753ZI microcontroller. Using the open
14 *Lithium Iron Phosphate (LFP) Cycle Ageing Dataset*, full-precision models pro-
15 vide the accuracy reference against which the compressed variants are evaluated.
16 Long-horizon replay yields mean absolute error (MAE) values of 2.3–2.8 per-
17 centage points for SOC and 0.9–1.5 for SOH, showing that compression largely
18 preserves accuracy. Hardware-in-the-loop benchmarks show that the evaluated
19 30 % pruning configuration reduces flash by 40.9 % and 45.5 % and inference
20 time by 42.9 % and 44.0 % for the SOC and SOH models, respectively. The
21 analytical complexity model links these savings to pruning fraction and network
22 dimensions. Quantization further reduces flash by 50.2 % and 58.8 %, but raises
23 runtime in the mixed-precision implementation. Together, the study provides a
24 reproducible reference for resource-efficient embedded battery-state estimation.
25 *Keywords:* Embedded artificial intelligence, Battery state estimation, Long
26 short-term memory network, Structured pruning, Weight quantization,
27 Microcontroller deployment

28 Nomenclature

29 *Acronyms*

30	BMS	Battery management system
31	SOC	State of charge
32	SOH	State of health
33	DoE	Design of Experiments
34	LFP	Lithium iron phosphate
35	LSTM	Long short-term memory network
36	MLP	Multi-layer perceptron
37	STM32	32-bit ARM Cortex-M microcontroller family
38	MAE	Mean absolute error
39	RMSE	Root-mean-square error
40	MAC	Multiply–accumulate operation
41	INT8	Signed 8-bit integer representation (quantized weights)
42	FP32	32-bit floating-point representation

43 *Symbols*

44	t	Time
45	$U(t)$	Terminal voltage at time t
46	$I(t)$	Current at time t (positive for discharge)
47	$T(t)$	Cell temperature at time t
48	$\text{EFC}(t)$	Equivalent full cycles accumulated up to t
49	$Q_c(t)$	Cumulative charge throughput up to t
50	C_{nom}	Nominal discharge capacity of the cell
51	C_{app}	Apparent capacity inferred from the latest check-up
52	DoD	Depth of discharge

53	$\text{SOC}(t)$	State of charge at time t
54	SOH_k	State of health at reference cycle k
55	$\mathbf{x}_t$	Input feature vector at time t
56	$\mathbf{h}_t$	LSTM hidden state at time t
57	$\mathbf{c}_t$	LSTM cell state at time t
58	$\mathbf{z}_t$	Concatenation of input and previous hidden state at time t
59	$\mathbf{i}_t, \mathbf{f}_t, \mathbf{o}_t, \mathbf{g}_t$	Input, forget, output, and candidate gates at time t
60	$\sigma(\cdot)$	Logistic sigmoid activation function
61	$\tanh(\cdot)$	Hyperbolic tangent activation function
62	$\odot$	Element-wise (Hadamard) product
63	W	Weight matrix (LSTM or MLP)
64	b	Bias vector
65	q	Quantized weight (INT8 code)
66	s	Per-row quantization scale (FP32)
67	$\hat{w}$	Reconstructed floating-point weight after dequantization
68	ε_q	Positive lower bound for the quantization scale
69	$\mathcal{R}$	Index set of retained LSTM hidden units
70	e_n	Signed prediction error of sample n in percentage points
71	$Q_p(\cdot)$	Empirical quantile operator at probability p
72	$W_{\text{ih}}^{\text{int8}}$	Quantized input-to-gate weight matrix
73	$W_{\text{hh}}^{\text{int8}}$	Quantized hidden-to-gate weight matrix
74	$s_{\text{ih}}, s_{\text{hh}}$	Per-row quantization scale vectors for $W_{\text{ih}}^{\text{int8}}$ and $W_{\text{hh}}^{\text{int8}}$
75	b_{fp32}	Floating-point bias of a given gate
76	g	Gate pre-activation used in the quantized LSTM kernel
77	D, H, M	Input dimension, LSTM hidden size, and MLP width
78	N_{MAC}	Analytical multiply-accumulate count per inference

79 1. Introduction

80 Accurate online estimation of the state of charge (SOC) and the state of
81 health (SOH) remains a central challenge for battery-powered systems [1, 2, 3, 4].
82 Recent reviews show sustained progress in model-based and data-driven estima-
83 tors, but they also emphasise that accuracy depends strongly on cell chemistry,
84 operating conditions, reference construction, and evaluation protocol [5, 1]. For
85 an embedded battery management system, predictive accuracy alone is there-
86 fore insufficient: the estimator must also fit the available flash and RAM, meet
87 the update deadline, and preserve its state consistently over long input streams.

88 The study is motivated by modular stationary storage systems in remote
89 microgrids. Within the MG Farm programme [6, 7, 8, 9], partners from France,
90 Algeria, Morocco, and Germany develop smart-grid demonstrators for North
91 African sites. The Technical University of Berlin contributes the battery ageing
92 experiments, the LFP DoE Cycle Ageing Dataset, a battery management sys-
93 tem prototype, and the SOC/SOH estimators deployed on its controller. Each
94 storage module must operate autonomously within the flash, RAM, and timing
95 limits of an STM32H753ZI-class microcontroller. Similar constraints arise in
96 mobile systems that cannot rely on continuous cloud access.

97 Existing embedded studies demonstrate that compact neural SOC or SOH
98 estimators are feasible on microcontrollers and that quantization can reduce
99 their storage requirements [10, 11, 12, 13, 14]. However, these studies generally
100 examine one battery state, one model family, or one deployment platform. They
101 do not provide a common, auditable comparison of structured hidden-unit prun-
102 ing and recurrent-weight quantization for both stateful SOC and SOH estima-
103 tors under identical long-horizon replay and hardware measurement conditions.
104 Reproduction is further hindered when scaling parameters, recurrent-state han-
105 dling, and the exact mixed-precision boundary are not reported.

106 This work uses the Lithium Iron Phosphate (LFP) DoE Cycle Ageing Dataset [15].
107 It contains long-term cyclic ageing measurements from multiple LFP cells under
108 full and fractional factorial conditions, with controlled temperature, charge and

109 discharge rates, depth-of-discharge variations, and interleaved capacity, open-
 110 circuit-voltage, and pulse-power check-ups. These data support a paired assess-
 111 ment of rapid SOC dynamics and slowly evolving SOH trajectories.

112 The objective of this study is to quantify, under identical data and deploy-
 113 ment conditions, how structured hidden-unit pruning and recurrent-weight post-
 114 training quantization change the accuracy, memory footprint, and inference cost
 115 of stateful LSTM + MLP SOC and SOH estimators on an STM32H753ZI. The
 116 study makes three contributions:

- 117 (i) an auditable path from trained PyTorch models to stateful C inference,
 118 including explicit feature scaling, recurrent-state handling, pruning, and
 119 mixed-precision quantization.
- 120 (ii) a controlled Base–Pruned–Quantized comparison for both SOC and SOH
 121 using continuous replay, linked-firmware flash and RAM, host latency, and
 122 cycle-counter-measured kernel time.
- 123 (iii) an evidence-bounded trade-off analysis that connects architecture-level
 124 operation counts and implementation details to long-horizon accuracy,
 125 filtering, and application-dependent utility.

126 The novelty lies in this paired, transparent deployment and compression bench-
 127 mark rather than in proposing a new recurrent cell or compression algorithm.

128 2. Related Work and Motivation

129 2.1. SOC/SOH Estimation

130 Existing SOC/SOH estimation methods can be grouped into three fami-
 131 lies [2, 3, 4, 5]. First, *measurement-derived* techniques use Coulomb counting
 132 and open-circuit-voltage (OCV) look-up tables during rest periods. They pro-
 133 vide inexpensive proxies but suffer from drift, offset accumulation, and a strong
 134 dependence on initialisation and periodic recalibration. Second, *model-based*
 135 observers combine equivalent-circuit or electrochemical models with Kalman fil-
 136 tering and its variants. These approaches can offer physical interpretability but
 137 require careful parameter identification and are sensitive to model mismatch and

138 ageing-induced parameter drift. Third, *data-driven* methods employ machine-
 139 learning regressors, including LSTM networks, Transformers, and hybrid struc-
 140 tures, to map sequences of current, voltage, and temperature to SOC and SOH
 141 estimates. Key challenges for such models are robustness when operated un-
 142 der conditions that differ from the training data and the careful treatment of
 143 normalisation and state handling on embedded targets.

144 Recent hybrid estimators combine physical or statistical preprocessing with
 145 learned correction models. Wang et al. [16] couple online parameter identifi-
 146 cation with an adaptive radial-basis correction and differential support-vector
 147 machine for SOC estimation under dynamic vehicle-oriented tests. For whole-
 148 life capacity estimation, Wang et al. [17] combine singular filtering, Gaussian-
 149 process regression, and an LSTM to address fast ageing and varying currents.
 150 These studies broaden the operating conditions and model structures consid-
 151 ered for battery-state estimation. Their primary evidence concerns estimation
 152 accuracy, however. Neither reports a paired post-training pruning and quantiza-
 153 tion study with linked-firmware memory and recurrent microcontroller timing.
 154 They are therefore complementary to, rather than substitutes for, the deploy-
 155 ment benchmark developed here. Across these families the reported accuracy
 156 spans a wide range. Electrochemical or Equivalent Circuit Model (ECM)-based
 157 observers with carefully tuned filters often achieve SOC root-mean-square errors
 158 (RMSE) well below 1 % of full scale and sub-percent SOH capacity errors on lab-
 159 oratory test cycles [18, 19, 20, 21]. Recent sequence-model approaches based on
 160 N-BEATS and Bidirectional LSTMs (BiLSTMs) for sodium-ion cells report SOC
 161 Mean Absolute Error (MAE) between roughly 0.1 and 0.4 % [22, 23], whereas
 162 TinyML-style SOC estimators deployed on microcontrollers often operate in the
 163 1–3 % error regime [10, 11, 12]. These values provide context rather than a di-
 164 rect ranking because chemistry, data splits, target construction, drive cycles,
 165 and error definitions differ between studies. Kalman filters, particle filters, and
 166 equivalent-circuit-model observers therefore dominate the classical literature,
 167 yet their accuracy can degrade over long operating periods with pronounced
 168 current and temperature gradients [19, 21]. Recurrent neural networks, and in

particular LSTM-based regressors, address these limitations by integrating long
 temporal context and absorbing residual modelling errors [24, 21]. In this work
 we deliberately restrict the architecture search to a single-layer LSTM with a
 fully connected Multi-Layer Perceptron (MLP) output layer. LSTMs retain ex-
 plicit cell states with separate input, forget, and output gates, which made it
 straightforward to hand-code a numerically stable, stateful kernel in C and to
 reuse the same backbone for both SOC and SOH. Gated recurrent units (GRUs)
 or Transformer variants could offer alternative trade-offs between accuracy and
 cost, but exploring multiple architectures would dilute the focus on the embed-
 ded tool flow and is left as future work. Most publications on LSTM-based
 estimators maximise accuracy with complex architectures and rarely examine
 how the resulting models behave once deployed on constrained controllers. Even
 within the LSTM family, sliding-window inference is commonly employed be-
 cause it delivers high accuracy, but it also requires substantial RAM to buffer
 multiple past samples. A stateful LSTM that hands over its hidden and cell
 vectors step by step retains the temporal information without maintaining large
 windows and therefore forms the first optimisation step in our architecture.
 The associated handling of hidden states, feature scaling, and numerical stabil-
 ity is rarely described in the literature, which is why this work explicitly targets
 embedded SOC/SOH deployment.

2.2. Embedded Deployment Strategies

Commercial toolchains such as ONNX Runtime, TensorRT, and STM32Cube.AI
 support many neural-network deployment workflows [25, 26, 27]. For continu-
 ously stateful estimators, however, a generated binary does not always expose
 recurrent-state handling or the precise placement of mixed-precision scaling op-
 erations needed for an implementation audit. Naguib et al. [14] compare accu-
 racy, execution time, flash, and RAM for classical and neural SOC estimators
 on two NXP microcontrollers. Crocioni et al. [10] deploy compact recurrent ca-
 pacity estimators on STM32 targets and study 8-bit quantization. More recent
 TinyML studies evaluate quantized or hardware-accelerated SOC estimators

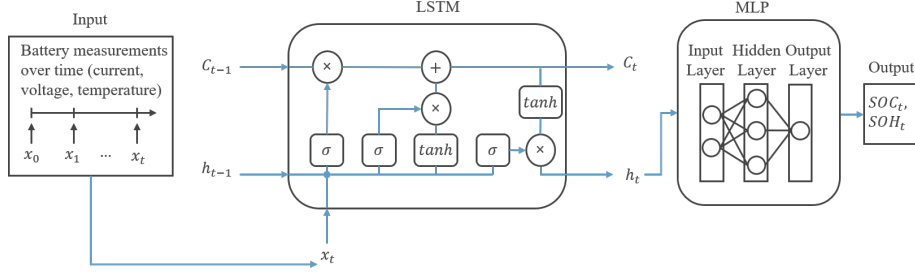


Figure 1: Schematic of the deployed SOC/SOH estimator. Preprocessed feature streams are fed into a single-layer LSTM whose hidden state is propagated across time. At each step a lightweight MLP head maps the hidden representation either to SOC or SOH, enabling stateful streaming inference on the STM32.

on embedded platforms [11, 12], while Kamali et al. [13] target computation-
ally compact SOH estimation. These studies establish embedded feasibility,
but each addresses a narrower combination of task, architecture, compression
method, and target hardware.

At the compression-method level, PQ Bench [28] systematically compares
pruning and quantization across general deep-learning workloads and confirms
that their accuracy-storage trade-offs are model dependent. It does not ad-
dress the continuous recurrent state, SOH post-processing, or microcontroller
timing behaviour of battery estimators. The resulting research gap is therefore
not the absence of compact neural estimators, but the absence of a controlled,
task-paired benchmark that traces two transparent compression routes from
trained SOC and SOH models to measured stateful firmware behaviour. Our
study addresses this gap with the same input dimension, streaming protocol,
performance-indicator definitions, and STM32 measurement procedure for Base,
Pruned, and Quantized variants of both tasks.

To retain control over the implementation boundary, we extract the Py-
Torch tensors, remove complete recurrent channels according to the documented
saliency score, apply per-row symmetric quantization to the recurrent weights,
and implement the LSTM gates and activations in C. This path makes the state

218 update, scaling operations, and stored parameter types inspectable and links the
 219 model transformations directly to the reported flash, RAM, and timing mea-
 220 surements.

221 **3. Dataset and Preprocessing**

222 *3.1. Ageing Experiments and Load Profiles*

223 All experiments rely on the high-resolution LFP DoE Cycle Ageing Dataset [15].
 224 The measurements originate from a design-of-experiments study on commer-
 225 cially available JGCFR18650-1800mAh-3.2V LFP cells [29] conducted at the
 226 Technical University of Berlin. Cells were cycled inside a climatic chamber at
 227 25°C while the following factors were systematically varied:

$$C_{\text{chg}} \in [0.5 \text{ C}, 0.9 \text{ C}], \quad (1)$$

$$C_{\text{dis}} \in [1.0 \text{ C}, 2.5 \text{ C}], \quad (2)$$

$$\text{DoD} \in [45\%, 65\%]. \quad (3)$$

228 The mixed full/fractional factorial plan distributes fifteen test cells across eight
 229 operating-point combinations and includes replication to assess repeatability.
 230 Figure 2 shows the low and high levels of all three factors.

231 Between ageing loops, dedicated check-ups provide capacity tests, open-
 232 circuit-voltage sweeps, and hybrid pulse-power characterisation. Stationary-
 233 storage profiles are additionally injected to emulate realistic power flows [30].
 234 These profiles originate from residential photovoltaic home-storage applications
 235 and contain stochastic load and generation patterns that excite the cell differ-
 236 ently from constant-current cycling.

237 The raw dataset contains uniformly sampled measurements at 1 Hz, includ-
 238 ing voltage, current, terminal temperature, and auxiliary channels. Partitioned
 239 Parquet tables support efficient retrieval of the long ageing sequences. The
 240 dataset is published in [15]. The cube notation follows standard DoE conven-
 241 tions as summarised in [31]. Together, Figures 3 and 4 illustrate the diversity of

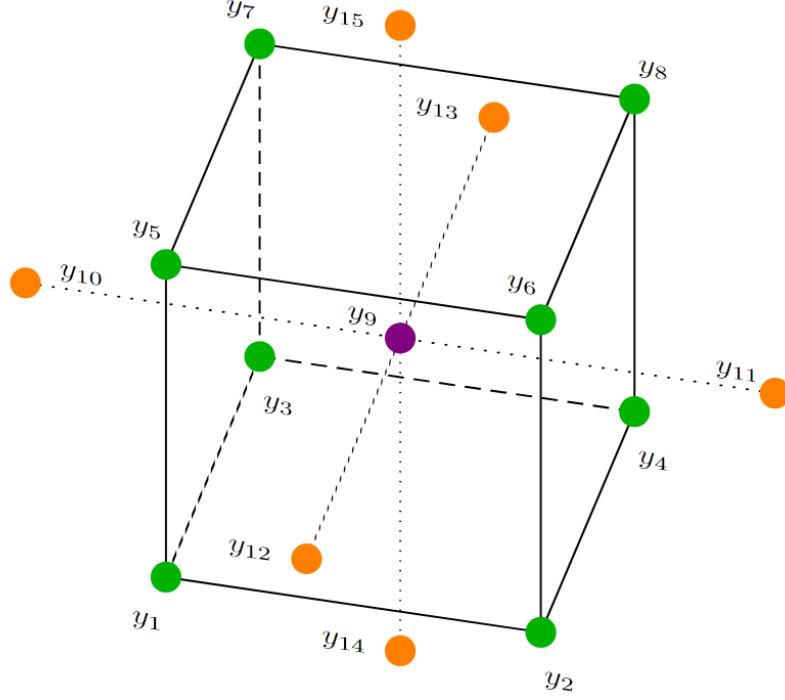


Figure 2: General three-factor design-of-experiments geometry used to structure the LFP ageing campaign. The cube represents the factorial design space, while the generic labels y_1 to y_{15} denote experimental conditions within and around it. The dataset-specific charge rate, discharge rate, and depth-of-discharge levels are described in the accompanying text.

242 SOC transients and ageing behaviour that the estimators must capture: short-
 243 term SOC dynamics are driven by fast current swings and mode changes between
 244 profiles, whereas SOH evolves slowly over hundreds of days and depends on the
 245 cumulative operating conditions of each cell.

246 3.2. Feature Engineering

247 The SOC model consumes the six-dimensional feature vector

$$\mathbf{x}_t^{\text{SOC}} = [U(t), I(t), T(t), Q_c(t), \frac{dU}{dt}(t), \frac{dI}{dt}(t)], \quad (4)$$

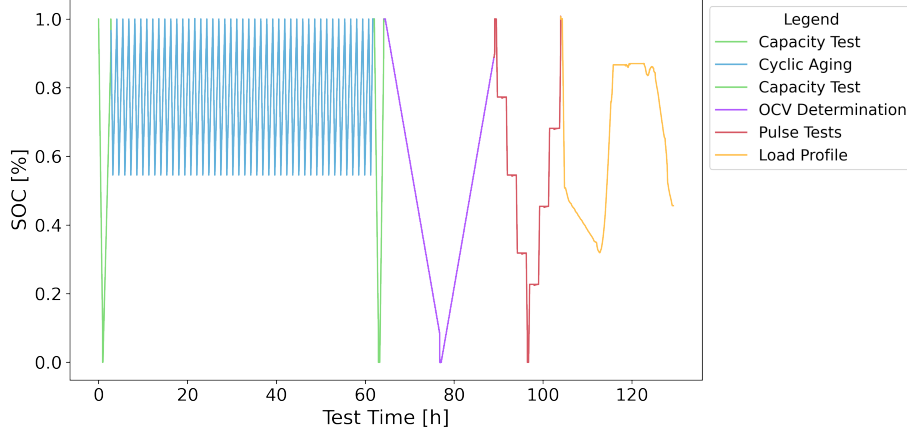


Figure 3: Representative SOC trajectory illustrating selected segments of the experimental protocol used in this study. The plot shows a portion of the cyclic ageing procedure with defined charge and discharge currents and a specified depth-of-discharge (DoD), followed by the diagnostic checkup sequence comprising OCV determination, HPPC pulses, and capacity tests. In addition, the trajectory includes a load profile taken from a stationary battery storage system. Together, these segments capture both controlled laboratory diagnostics and application-oriented operating conditions.

248 where the audited preprocessing uses timestamp-aware backward differences
 249 rather than centred differences. For sample index $k > 0$,

$$\Delta t_k = \max(t_k - t_{k-1}, \varepsilon_t), \quad \varepsilon_t = 10^{-6}, \quad (5)$$

$$\left. \frac{dU}{dt} \right|_k = \frac{U_k - U_{k-1}}{\Delta t_k}, \quad \left. \frac{dI}{dt} \right|_k = \frac{I_k - I_{k-1}}{\Delta t_k}. \quad (6)$$

250 This causal calculation requires only the preceding voltage/current sample and
 251 timestamp and has constant work per update. In the evaluated benchmark, how-
 252 ever, derivative and cumulative channels were prepared host-side before replay.
 253 Complete six-feature vectors were transmitted via UART, and the firmware
 254 applied the same robust-scaling parameters used during model development.
 255 Consequently, the reported DWT kernel times exclude sensor acquisition and
 256 feature construction. The exact placement of the six-feature scaling operation
 257 is documented with the STM32 measurement interval below. Each component
 258 is normalised using a robust scaling operation based on the median m and in-

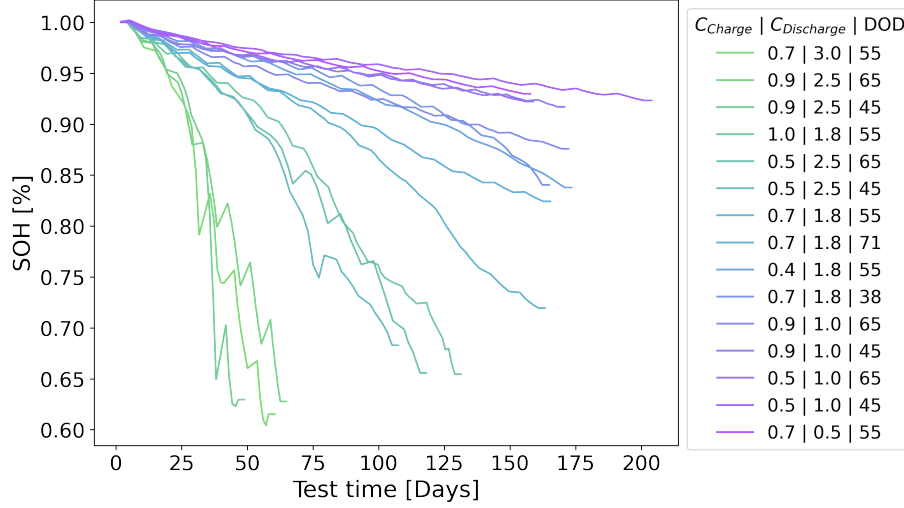


Figure 4: Evolution of the state of health across all LFP cells in the ageing campaign, shown over the total experimental duration. The spread between trajectories reflects the different operating points in the factorial plan (C-rates, depth of discharge, profile mix) and motivates the use of a data-driven SOH estimator that generalises across cells and loading conditions.

terquartile range r :

$$\tilde{x}_{t,k}^{\text{SOC}} = \frac{x_{t,k}^{\text{SOC}} - m_k}{r_k + \varepsilon}, \quad \varepsilon = 10^{-6}. \quad (7)$$

The same (m_k, r_k) pairs are embedded in the firmware so that host and STM32 inputs are normalised identically. The SOH model uses a slightly different feature stack to capture slow-varying degradation markers:

$$\mathbf{x}_t^{\text{SOH}} = [t, U(t), I(t), T(t), \text{EFC}(t), Q_c(t)], \quad (8)$$

where t is the cumulative time stamp and $\text{EFC}(t)$ denotes the equivalent full cycles accrued up to sample t . These channels are likewise robustly normalised, relying on median and interquartile range rather than mean and variance to reduce the influence of outliers in current and voltage measurements [32, 33]. During deployment, the SOC and SOH models run independently on the microcontroller and each produces one estimate per second. In principle, both estimates could be fused more tightly, but this work keeps the models separate to evaluate their resource use and accuracy independently.

271 3.3. Reference SOC and SOH targets

272 The ground-truth SOC trajectory is derived from Coulomb counting with
 273 drift compensation. Denoting the apparent cell capacity after a given check-up
 274 by C_{app} , the SOC inside a charge-discharge segment is computed as

$$\text{SOC}(t) = \text{SOC}(t_0) - \frac{1}{C_{\text{app}}} \int_{t_0}^t I(\tau) d\tau, \quad (9)$$

275 where the integral is reset to $\text{SOC} = 1$ after each constant-voltage (CV) ter-
 276 mination. The reset mitigates the bias that would otherwise accumulate due
 277 to capacity fade, yet the long-term traces still reveal gradual changes in the
 278 SOC envelope as C_{app} decreases. State of health is obtained from the periodic
 279 capacity check-ups through

$$\text{SOH}_k = \frac{C_k}{C_{\text{nom}}}, \quad (10)$$

280 with C_k representing the measured discharge capacity during the k -th reference
 281 cycle (comprising capacity test, open-circuit-voltage sweep, and hybrid pulse-
 282 power characterisation) relative to the nominal capacity C_{nom} . These targets
 283 highlight the need for a data-driven estimator that remains accurate despite
 284 the evolving capacity landscape, which motivates the use of recurrent neural
 285 networks for both SOC and SOH predictions. The reference SOC and SOH
 286 traces are therefore offline constructs. They rely on information that is not
 287 available in deployment, such as long rest periods for OCV stabilisation and
 288 capacity from periodic full cycles, and they assume perfect initialisation and
 289 recalibration. These constructions follow common practice in battery manage-
 290 ment, where SOC is obtained by Coulomb counting and SOH is expressed as the
 291 ratio between present and nominal capacity [34, 2, 3]. On the STM32 the esti-
 292 mators operate purely on the streamed measurements and their internal states.
 293 The reference trajectories only serve as ground truth for training and evaluation.

294 3.4. Streaming inference setup

295 Because the SOC and SOH estimators are stateful, the LSTM hidden and
 296 cell states are continuously forwarded from one sample to the next along the full

input stream. At each sampling instant the current feature vector is processed together with the previous hidden and cell states, which yields the new states and the corresponding SOC or SOH estimate. We therefore operate the models in a purely step-by-step streaming regime without forming sliding input windows. On the host the full normalised sequence from the LFP DoE Cycle Ageing Dataset is processed once in this fashion, and the resulting SOC and SOH estimates are aligned with the reference trajectories. On the STM32 firmware the same step-wise LSTM implementation is used, driven by the UART feature stream, so that host and embedded benchmarks reflect identical inference conditions. The streaming dashboards in Figs. 8 and 11 illustrate a complete run for one representative cell.

3.5. SOH prediction post-processing

The SOH trajectories in the final dashboard use two sequential causal filtering stages. First, each raw model output $\hat{y}_t$ is multiplicatively aligned with the initial SOH label, $\hat{y}_t^{(0)} = (y_0/\hat{y}_0)\hat{y}_t$. Stage 1 limits the change of this calibrated prediction relative to the preceding stage output to

$$\ell_t = \min\left(10^{-4}|y_{t-1}^{(1)}|, 10^{-5}\right), \quad (11)$$

and subsequently applies an exponential moving average (EMA) with $\alpha_1 = 0.02$. The limiter is symmetric, so positive and negative proposal changes are both restricted to $\pm\ell_t$. Stage 2 consumes the complete stage-1 trajectory, applies another EMA with $\alpha_2 = 10^{-6}$, and limits only downward changes according to

$$y_t^{(2)} = \max\left(e_t^{(2)}, y_{t-1}^{(2)} - 2 \times 10^{-8}\right), \quad e_t^{(2)} = (1 - \alpha_2)e_{t-1}^{(2)} + \alpha_2 y_t^{(1)}. \quad (12)$$

Thus, the curve in Fig. 11 is not obtained by applying α_2 directly to an unfiltered network output.

For an EMA sampled at interval Δt , the equivalent time constant and cutoff frequency are

$$\tau = -\frac{\Delta t}{\ln(1 - \alpha)}, \quad f_c = \frac{1}{2\pi\tau}. \quad (13)$$

At the dataset rate of 1 Hz, stage 1 has $\tau = 49.5$ s, $f_c = 3.22$ mHz, and an EMA-only 90 % step-response time of approximately 114 s. For stage 2, τ is approximately 11.57 days, $f_c = 1.59 \times 10^{-7}$ Hz, and the corresponding 90 % time is 26.65 days. These frequencies characterise only the linear EMA components. The nonlinear rate limiters have no single cutoff frequency. The very strong second stage can reduce error against slowly varying labels while introducing substantial response delay, which is evaluated separately in Appendix A.3.

4. Model Architecture and Baseline Performance

4.1. LSTM architecture in PyTorch

Both SOC and SOH estimators employ the same recurrent equation set but differ in size according to their training releases. The SOC configuration uses $H = 64$ hidden units and an MLP output layer of width 64, whereas the SOH configuration increases both dimensions to 128 to cope with the longer temporal context. LSTM gates i_t, f_t, o_t and candidate state g_t follow the standard formulation used in modern RNN surveys [35]. All models are implemented in PyTorch [36]. For one streaming update with input dimension $D = 6$, LSTM hidden size H , and fixed MLP width M , the matrix products require

$$N_{\text{MAC}}(H) = 4H(D + H) + HM + M, \quad (14)$$

where one multiply-accumulate (MAC) is counted per weight. Bias additions, nonlinearities, indexing, and memory transfers are excluded. For a hidden-unit pruning fraction p , the retained hidden size is $H_p = (1 - p)H$. Substitution into Eq. (14) gives the relative analytical MAC reduction

$$\begin{aligned} R_{\text{MAC}}(H, p) &= 1 - \frac{N_{\text{MAC}}((1 - p)H)}{N_{\text{MAC}}(H)} \\ &= \frac{4(2p - p^2)H^2 + p(4D + M)H}{4H^2 + (4D + M)H + M}, \\ \lim_{H \rightarrow \infty} R_{\text{MAC}}(H, p) &= 2p - p^2. \end{aligned}$$

342 The quadratic recurrent term is therefore reduced by $2p - p^2$, the linear input
 343 and MLP terms by p , and the fixed output term is unchanged. The finite-
 344 model reduction is a weighted combination of these contributions and remains
 345 below the large- H limit. For the evaluated fraction $p = 0.30$, this limit is
 346 51 %. The implemented integer dimensions give 22,080 to 12,124 MACs for
 347 SOC ($64 \rightarrow 45$), a 45.1 % reduction, and 85,120 to 46,208 MACs for SOH
 348 ($128 \rightarrow 90$), a 45.7 % reduction. Appendix A.6 visualises both this operating
 349 point and the general dependence on p . The relation makes the computational
 350 effect of a selected hidden-size reduction predictable before deployment, while
 351 hardware measurements translate the analytical reduction into platform-specific
 352 flash, latency, and energy changes.

$$\begin{aligned}
 \mathbf{z}_t &= [\mathbf{x}_t, \mathbf{h}_{t-1}], \\
 \mathbf{i}_t &= \sigma(W_i \mathbf{z}_t + \mathbf{b}_i), \quad \mathbf{f}_t = \sigma(W_f \mathbf{z}_t + \mathbf{b}_f), \\
 \mathbf{o}_t &= \sigma(W_o \mathbf{z}_t + \mathbf{b}_o), \quad \mathbf{g}_t = \tanh(W_g \mathbf{z}_t + \mathbf{b}_g), \\
 \mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t, \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t).
 \end{aligned} \tag{15}$$

353 Equations (14) and the recurrent update above define the common Base ar-
 354 chitecture used for both tasks. Pruning changes its hidden dimension and pa-
 355 rameter submatrices, whereas quantization changes the stored representation of
 356 selected recurrent weights. Neither transformation changes the input features,
 357 targets, or streaming state-update order.

358 Both SOC and SOH models are trained with a mean-squared-error loss be-
 359 tween the prediction and the corresponding reference at each training sample.
 360 During validation we additionally report mean absolute error (MAE), root-
 361 mean-square error (RMSE), and R^2 . The trained parameters, configuration
 362 files, and optimiser statistics are retained to enable deterministic replay of both
 363 models.

364 4.2. Training Notes

365 Training is carried out on a GPU node equipped with NVIDIA A100-class
 366 accelerators using mixed-precision (FP16) arithmetic. Both models use the

AdamW optimiser with cosine-annealing warm-restart scheduling, gradient clipping, and mini-batch training [37, 38, 39, 40]. The SOC configuration uses a learning rate of 1.5×10^{-4} , a batch size of 512, and sequence chunks of length 2024. The corresponding SOH values are 2.0×10^{-4} , 128, and 2048.

Deployment operates the LSTM in stateful, step-by-step streaming mode, whereas training uses truncated backpropagation through time over these chunks. Hidden and cell states are propagated within every chunk in the same temporal order as during deployment. Chunking permits randomised segment sampling and tractable gradient computation without changing the recurrent update equations.

4.3. Baseline Accuracy

On the workstation the SOC and SOH base models are first evaluated in streaming mode on the full ageing trajectory of one representative LFP cell from the LFP DoE Cycle Ageing Dataset [15], yielding baseline MAE and RMSE values for both tasks. These Python replays operate on the original PyTorch implementations and provide the numerical reference for the compression experiments. The corresponding trained models are pruned, quantized, and exported to STM32 firmware, where identical feature streams are replayed to measure accuracy, flash and RAM usage, system latency, and on-device inference time. A timing-derived energy proxy is calculated as described in Section 7. Figure 5 summarises this training-to-firmware benchmarking pipeline.

5. Pruning

Pruning reduces the size and computational cost of a neural network by eliminating parameters or whole units that contribute little to the overall mapping, while keeping the input-output behaviour as close as possible to the dense baseline [41, 42]. In our setting we apply structured pruning at the level of complete LSTM hidden units so that the resulting model remains well suited for efficient C implementations on the microcontroller.

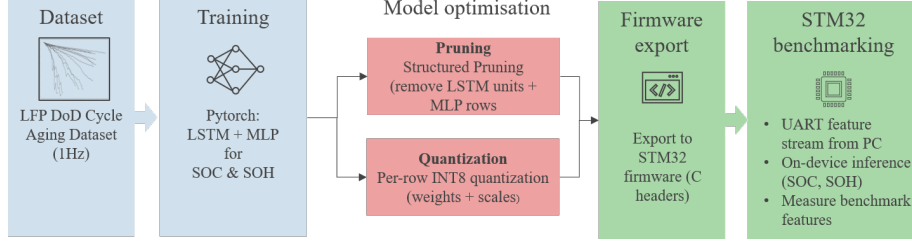


Figure 5: End-to-end pipeline from dataset to embedded benchmarking. The LFP DoE Cycle Ageing Dataset feeds PyTorch training on a GPU workstation. The resulting SOC and SOH LSTM+MLP models are pruned and quantized, exported as C headers, and deployed on an STM32H753ZI board, where flash/RAM usage, inference time, and latency are measured under streaming replay. Estimated energy per inference is derived from kernel time under a constant board-power assumption.

5.1. Structured magnitude pruning

Structured magnitude pruning removes the lowest-saliency LSTM units until only 45 hidden channels remain in the SOC model (30% sparsity), with analogous settings for the SOH experiments. Reducing the state dimension is a prerequisite for fitting the estimators into the flash and SRAM budgets of the target STM32-based BMS and for shortening the on-device inference time, inspired by structured sparsity approaches for deep and recurrent networks [43, 44, 45]. In analogy to the group-lasso style regularisers proposed for LSTMs in [43, 44], which aggregate weights across gates via L2 norms, we define a saliency score for each hidden unit h that combines all incoming weights from the different gates:

$$s_h = \sum_{g \in \{i, f, o, g\}} \left(\|W_{ih}^{(g)}[h, :]\|_2 + \|W_{hh}^{(g)}[h, :]\|_2 \right). \quad (16)$$

Units are ranked by s_h , and the H' highest-scoring indices form the retained set $\mathcal{R}$. For each gate g , structured pruning constructs the reduced matrices

$$\begin{aligned} \widetilde{W}_{ih}^{(g)} &= W_{ih}^{(g)}[\mathcal{R}, :], \\ \widetilde{W}_{hh}^{(g)} &= W_{hh}^{(g)}[\mathcal{R}, \mathcal{R}], \\ \widetilde{W}_{MLP,1} &= W_{MLP,1}[:, \mathcal{R}]. \end{aligned} \quad (17)$$

408 Thus the same index set removes gate rows, recurrent columns, and the cor-
 409 responding inputs of the first MLP layer. The reduced tensors remain dense.
 410 The computational saving comes from the smaller dimension rather than sparse
 411 indexing.

412 The final SOC deployment uses $H' = 45$ instead of 64 hidden units, while
 413 SOH uses $H' = 90$ instead of 128. Both transformations remove approximately
 414 30 % of the recurrent channels. The fraction $p = 0.30$ was selected as a sin-
 415 gle moderate operating point that provides a substantial dimensional reduction
 416 while retaining useful estimator accuracy. It was not obtained from an ex-
 417 haustive pruning-rate sweep and is not claimed to be optimal. Establishing
 418 an optimal fraction would require separately pruned and fine-tuned models for
 419 each candidate rate and is reserved for a pruning-focused investigation. This
 420 is a one-shot structured operation followed by brief post-pruning fine-tuning.
 421 No iterative pruning schedule or pruning-aware training is used. The L2 ag-
 422 gregation gives a gate-consistent ranking that can be calculated directly from
 423 the model weights and mapped to dense C arrays. Appendix A.1 reports the
 424 resulting SOC saliency ranking and recurrent-weight distributions. These di-
 425 agnostics describe the resulting weight selection but do not establish a causal
 426 regularisation effect. The criterion is selected because it combines all four gate
 427 rows for one recurrent unit and maps directly to removal of dense rows, re-
 428 current columns, and the associated MLP input. It therefore provides a re-
 429 producible, gate-consistent ranking that translates directly into a smaller dense
 430 kernel. Gradient sensitivity would additionally use training data and backprop-
 431 agation, while activation-based ranking would use a representative calibration
 432 stream and a temporal aggregation rule. A future comparison with these data-
 433 dependent alternatives can determine whether their additional information fur-
 434 ther improves the accuracy–efficiency trade-off. Appendix A.7 summarises the
 435 method properties.

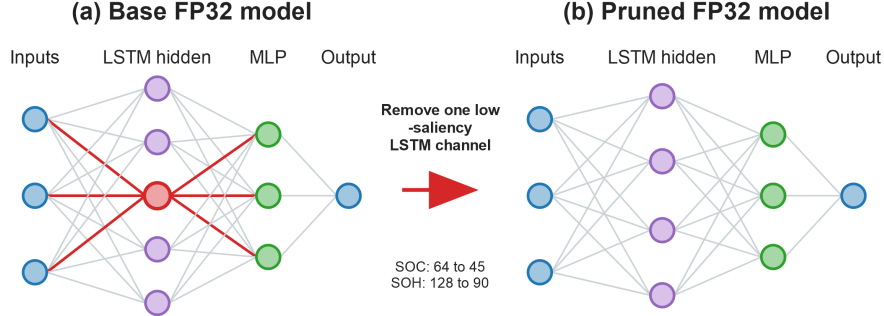


Figure 6: Structured hidden-channel pruning used in this study. Panel (a) marks one low-saliency LSTM channel and its incident connections in red. Panel (b) shows the resulting dense network after removing the corresponding gate rows, recurrent columns, and MLP input column according to Eq. (17). The final SOC and SOH hidden dimensions are shown between the panels.

5.2. Resulting pruned models

The resulting pruned SOC and SOH models still run in FP32 but reduce flash consumption and runtime considerably once compiled for STM32. Section 7 summarises the corresponding latency and memory savings. These pruned variants are evaluated alongside their base and quantized counterparts throughout the results section.

6. Quantization

Quantization reduces the numerical precision of selected weights while keeping the network topology unchanged. We apply post-training, weight-only quantization to the LSTM input-to-hidden and hidden-to-hidden matrices. Biases, activations, recurrent hidden and cell states, and the MLP remain FP32. The resulting firmware is therefore a mixed-precision storage implementation, not a fully integer inference path. Quantization-aware training is not used.

449 *6.1. Per-row symmetric quantization*

450 For row r of either recurrent matrix, the implementation computes

$$s_r = \max\left(\frac{\max_h |w_{r,h}|}{127}, \varepsilon_q\right), \quad (18)$$

$$q_{r,h} = \text{clip}\left(\text{round}\left(\frac{w_{r,h}}{s_r}\right), -127, 127\right), \quad (19)$$

$$\hat{w}_{r,h} = s_r q_{r,h}, \quad (20)$$

451 where ε_q prevents division by zero for an all-zero row. Although the INT8
 452 storage type provides 256 machine codes, the implementation deliberately leaves
 453 -128 unused and employs the symmetric 255-level set $\{-127, \dots, 127\}$. This
 454 keeps zero exactly representable and assigns equal positive and negative ranges.
 455 With the max-absolute scale in Eq. (18), finite row weights do not clip, and
 456 nearest-integer rounding gives the element-wise reconstruction bound

$$|w_{r,h} - \hat{w}_{r,h}| \leq \frac{s_r}{2}. \quad (21)$$

457 Per-row scaling gives small-magnitude rows a finer step size than one global
 458 scale, following standard uniform post-training quantization practice [46, 47].

459 *6.2. Mixed-precision gate computation*

460 Let r denote one gate row. The firmware evaluates

$$g_r = \sum_{d=1}^D x_d (s_{ih,r} q_{ih,r,d}) + \sum_{j=1}^H h_j (s_{hh,r} q_{hh,r,j}) + b_{fp32,r}. \quad (22)$$

461 Thus the recurrent weights are stored as INT8 codes but reconstructed into
 462 FP32 products inside the current accumulation loops. A built-in streaming
 463 comparison validates the exported gate computations against the source model
 464 and typically yields MAE differences below 10^{-4} .

465 The recurrent matrices account for approximately 80 % of the raw Base
 466 model constants in both tasks. Targeting these matrices concentrates the stor-
 467 age reduction on the dominant constants while retaining FP32 biases, activa-
 468 tions, recurrent states, and the smaller MLP path. This creates a transparent

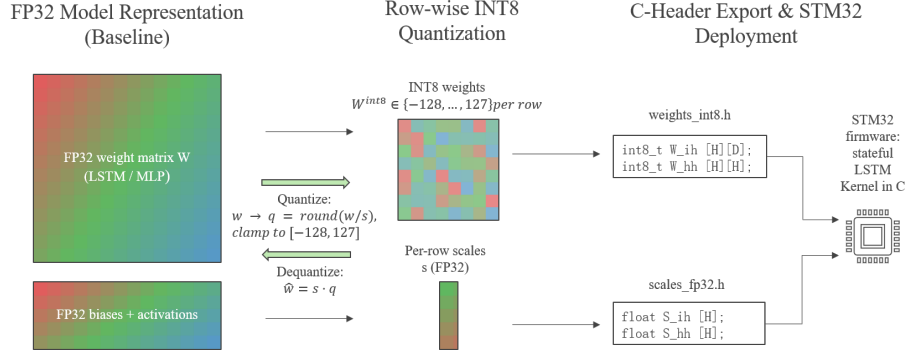


Figure 7: FP32-to-INT8 quantization and embedded deployment. The recurrent FP32 weight matrices are mapped row by row to INT8 codes and one FP32 scale per row. The INT8 storage type spans $\{-128, \dots, 127\}$, while the implemented symmetric quantizer generates $\{-127, \dots, 127\}$ and leaves -128 unused. The integer recurrent weights and scale vectors are exported as C headers and combined with FP32 biases, activations, recurrent states, and MLP operations in the STM32 kernel.

469 mixed-precision configuration that reduces persistent storage without changing
 470 the recurrent state representation. Quantizing activations and states could ex-
 471 tend the approach toward additional RAM and latency reductions and is consid-
 472 ered as a subsequent optimisation step. Appendix A.8 documents the precision
 473 boundary and model-constant composition.

474 6.3. Export for STM32

475 The exported headers are integrated into separate C firmware projects for
 476 the stateful SOC and SOH estimators. Each project combines the LSTM and
 477 MLP kernels with UART telemetry. A small driver library provides build files
 478 and flashing scripts so that identical measurement binaries can be rebuilt on
 479 other machines. This explicit tool chain, from training through pruning and
 480 quantization to firmware, exposes the numerical operations and records the
 481 build settings used for the resource measurements in Section 7.

482 7. Benchmark Methodology

483 This section describes how we benchmark the SOC and SOH models on both
484 the host PC and an STM32 Nucleo-144 development board with an STM32H753ZI
485 MCU [48, 14]. The microcontroller provides 2 MB of on-chip Flash and 1 MB of
486 SRAM, which constitute the budget for firmware code, model parameters, and
487 runtime buffers. Starting from the trained LSTM + MLP models, we generate
488 three embedded variants (full-precision baseline, pruned model, and quantized
489 model) and evaluate them on the *LFP DoE Cycle Ageing Dataset*. The bench-
490 mark combines long-horizon streaming simulations with hardware-in-the-loop
491 measurements and is fully scripted to ensure reproducible accuracy and resource
492 evaluations.

493 On the host side, the complete reference ageing sequence is replayed in
494 Python using the same streaming inference functions that are later exported
495 to C. The SOC and SOH estimators are evaluated in continuous time without
496 resetting their recurrent states, and error metrics (MAE, RMSE, R^2), error his-
497 tograms, and parity plots are computed from the resulting trajectories. These
498 simulations provide a high-precision reference for the embedded experiments
499 and are used to assess how pruning and quantization affect the dynamics of the
500 baseline models. To test whether deviations grow over time, the full-stream tra-
501 jectories are divided into ten consecutive, equal-count segments. Target MAE
502 and P95 error are computed in every segment, together with the mean abso-
503 lute deviation of each compressed variant from the Base output. Appendix A.2
504 reports this temporal analysis.

505 For the embedded benchmarks the base, pruned, and quantized SOC/SOH
506 models are embedded into dedicated firmware projects for the STM32H753ZI.
507 Because STM32Cube AI does not natively support stateful LSTM layers, the
508 recurrent update equations, post-processing, and UART diagnostics are imple-
509 mented manually in C. Identical stimulus streams are then injected via UART,
510 allowing a direct comparison between host-side and on-device behaviour under
511 the same operating conditions.

512 7.1. Limited software-level fault sensitivity

513 Two limited software-level sensitivity analyses examine the Base, Pruned,
 514 and Quantized variants under missed reports and transient input corruption.
 515 First, randomly omitted outputs at rates of 1, 5, and 10 %, as well as ten
 516 deterministic gaps of 60, 600, or 3600 samples, are replaced by the preceding
 517 prediction using a hold-last strategy.

518 The second analysis evaluates how a one-sample input-buffer fault propa-
 519 gates through the recurrent state. Each exported C model processes 8000 con-
 520 secutive C07 samples at 1 Hz. Thirty independent events are distributed over
 521 this sequence after at least 2048 preceding clean samples, with ten events each
 522 for voltage, current, and temperature. At every event, the disturbed branch
 523 starts from the same recurrent state as the clean branch. Bit 22, the most
 524 significant FP32 mantissa bit, is flipped in one input value before inference for
 525 exactly one sample. The following 60 inputs are unmodified and the recurrent
 526 state is not reset.

527 For event e and offset $k = 0, \dots, 60$, the fault-induced deviation is $d_{e,k} =$
 528 $100|\hat{y}_{e,k}^{\text{fault}} - \hat{y}_{e,k}^{\text{clean}}|$ in percentage points. Peak deviation and residual deviation
 529 after 60 s quantify the transient response independently of the model’s pre-
 530 existing target error. Accuracy is evaluated over the same 61-sample window
 531 using

$$\Delta\text{MAE}_{61,e} = \frac{100}{61} \sum_{k=0}^{60} (|\hat{y}_{e,k}^{\text{fault}} - y_{e,k}| - |\hat{y}_{e,k}^{\text{clean}} - y_{e,k}|). \quad (23)$$

532 A positive value indicates that the bit flip increases target MAE, while a neg-
 533 ative value means that the disturbed prediction happens to lie closer to the
 534 target. Recovery is the first time after the peak at which $d_{e,k}$ remains below
 535 $\max(0.1d_{e,\text{peak}}, 10^{-4} \text{ pp})$ for five consecutive samples. This test represents a
 536 transient input-buffer upset in software. It does not replace hardware fault
 537 injection into weights, activations, recurrent-state memory, communication, or
 538 MCU memory.

539 7.2. STM32 Measurements

540 On the STM32 Nucleo-144 development board, dedicated SOC and SOH
 541 firmware images execute the embedded estimators while host-side scripts handle
 542 stimulus injection and UART logging. Recorded feature streams are transmit-
 543 ted to the microcontroller, which runs the stateful LSTM + MLP and returns
 544 predictions with timing diagnostics.

545 Each sample produces two time stamps. Host latency measures the delay
 546 between sending an input vector and receiving its estimate. On-device inference
 547 time is measured with the data watchpoint and trace (DWT) cycle counter
 548 around the LSTM and MLP kernels. Reported values are averages over 10,000
 549 streaming inferences. The timed interval includes instructions and memory
 550 accesses executed by these kernels but excludes UART transfer, host processing,
 551 and host-side feature construction.

552 In the SOC projects, six-feature robust scaling lies inside the timed estimator
 553 call. The SOH Base and Pruned projects apply scaling immediately before the
 554 DWT interval, whereas the Quantized SOH forward function includes it. This
 555 small affine preprocessing cost was not profiled separately and is absent from
 556 the static operation counts. All inspected benchmark projects were compiled
 557 with GCC at -O0, using the hard-float ABI and Cortex-M7 FPUv5 floating-point
 558 unit. The measurements therefore characterise auditable, unoptimised reference
 559 kernels rather than production-optimised integer deployments.

560 7.3. Key Performance Indicators

561 For normalised target y_n and prediction $\hat{y}_n$, the signed error is expressed in
 562 percentage points as $e_n = 100(\hat{y}_n - y_n)$. The reported accuracy indicators are

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |e_n|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N e_n^2}, \quad (24)$$

$$\text{P95} = Q_{0.95}(\{|e_n|\}_{n=1}^N), \quad (25)$$

563 where $Q_{0.95}$ is the empirical 95th percentile. MAE and RMSE use all samples,
 564 whereas P95 identifies the absolute-error threshold below which 95 % of samples

Table 1: Engineering KPIs for SOC models on STM32H753ZI (full-stream averages). Energy is estimated from measured inference time under an assumed constant board power of 0.5 W.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	E_{est} [mJ]
Base FP32	12.70	1.40	105.32	4.93	0.70
Pruned FP32	12.20 (-3.9%)	0.80 (-42.9%)	62.27 (-40.9%)	4.03 (-18.3%)	0.40 (-42.9%)
Quant INT8	18.48 (+45.5%)	6.99 (+399%)	52.48 (-50.2%)	3.96 (-19.7%)	3.49 (+399%)

lie.

For each deployment we additionally report average host latency, on-device inference time, flash and SRAM usage, and an estimated energy-per-inference proxy. Tables 1 and 2 focus on these STM32 resource and timing indicators. Latency is the end-to-end UART delay, whereas inference time covers only the DWT-measured kernel interval. RAM combines statically allocated data and BSS sections with maximum observed streaming stack usage.

Energy was not measured separately for each model. Instead, we use

$$E_{\text{est}} = P_{\text{assumed}} t_{\text{inf}}, \quad P_{\text{assumed}} = 0.5 \text{ W}, \quad (26)$$

where 0.5 W is a representative average-power assumption for the complete development board. Consequently, the percentage change in E_{est} is identical to the percentage change in inference time. The proxy does not separate core computation, flash access, SRAM access, voltage-regulator losses, or other board consumers and cannot determine whether flash access dominates the actual energy consumption. An analogous set of measurements was recorded for the SOH firmware deployments. Table 2 lists the corresponding indicators.

The KPIs in Tables 1 and 2 highlight the trade-offs introduced by pruning and quantization. For SOC, pruning reduces the flash footprint by roughly 41 % and inference time by about 43 %, while latency remains dominated by UART transfer. Quantization approximately halves flash usage, but per-weight scaling and integer–float conversions in the mixed-precision kernels increase on-device inference time by a factor of four relative to Base. The SOH deployments show

Table 2: Engineering KPIs for SOH models on STM32H753ZI (full-stream averages). Energy is estimated from measured inference time under an assumed constant board power of 0.5 W.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	E_{est} [mJ]
Base FP32	34.88	22.73	335.00	8.69	11.37
Pruned FP32	24.69 (-29.2%)	12.72 (-44.0%)	182.41 (-45.5%)	6.96 (-19.9%)	6.36 (-44.0%)
Quant INT8	41.23 (+18.2%)	29.21 (+28.5%)	138.00 (-58.8%)	6.70 (-22.9%)	14.60 (+28.5%)

the same qualitative behaviour. Pruning saves about 46 % of flash and 44 % of inference time, whereas Quantized attains the smallest flash footprint but incurs a 29 % timing increase. Under the constant-power assumption, the estimated energy proxy follows these timing changes exactly. It is not independent evidence of board-level energy efficiency.

8. Results

The evaluation compares the full-precision Base, Pruned, and Quantized SOC and SOH estimators under the same continuous streaming replay of the LFP DoE Cycle Ageing Dataset. The following sections separate prediction accuracy, temporal behaviour, embedded resource use, and the resulting multi-objective trade-offs.

8.1. Streaming Prediction Accuracy

The primary metric for any battery state estimator is its ability to track the ground truth over extended periods without divergence. Figures 8 and 11 illustrate the streaming performance of the SOC and SOH models over the complete test duration. Across the complete SOC replay, the Base, Pruned, and Quantized variants obtain MAE values of 2.685, 2.338, and 2.791 percentage points, respectively. The Quantized trace often overlaps the Base trace visually, whereas the Pruned model has the lowest aggregate target MAE despite a larger model-to-Base deviation in some parts of the sequence. These results confirm that

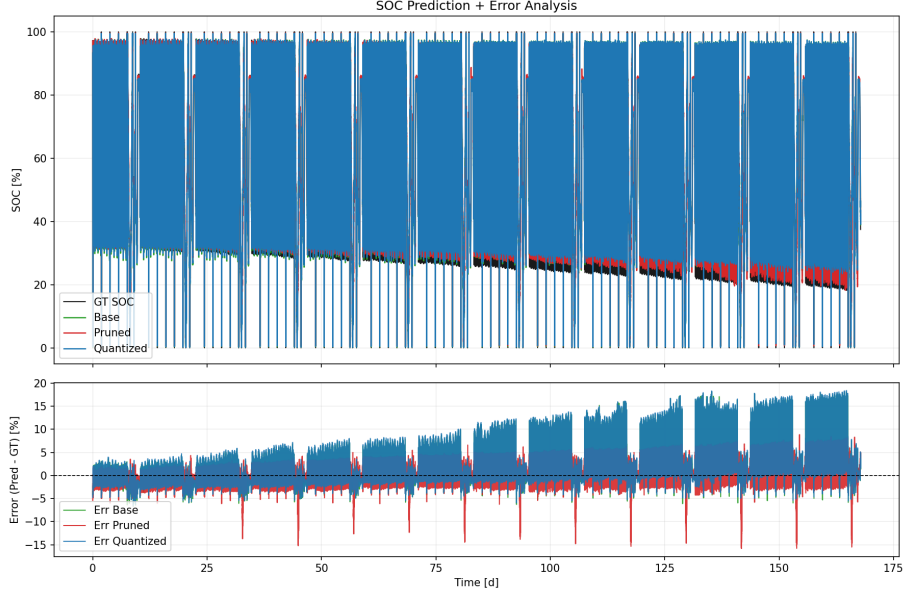


Figure 8: Streaming SOC dashboard on the full ageing trajectory. The black line represents the reference SOC, while the coloured lines show the predictions of the Base (green), Pruned (red), and Quantized (blue) models. The lower sub-panel displays the instantaneous prediction error.

606 post-training weight quantization preserves useful recurrent behaviour through-
 607 out the evaluated stream. Replays of additional operating sequences can extend
 608 this comparison to a wider range of temporal conditions.

609 The SOH estimator faces a different challenge: it must infer slow capacity
 610 degradation from sparse check-up events and continuous monitoring. In the
 611 plotted trajectories the discrete capacity tests have been removed from the esti-
 612 mator input, while the remaining check-up blocks (OCV sweeps, HPPC pulses,
 613 and profile segments) are retained. The plateau-like sections of the reference
 614 curve therefore correspond to these diagnostic intervals. The dashboard applies
 615 the complete two-stage causal post-processing chain from Section 3.5: initial
 616 alignment, stage 1 with $\alpha_1 = 0.02$ and a symmetric limiter, followed by stage 2
 617 with $\alpha_2 = 10^{-6}$ and a downward limiter. The resulting smoothing suppresses
 618 short-term variation but introduces a substantial accuracy-delay trade-off.

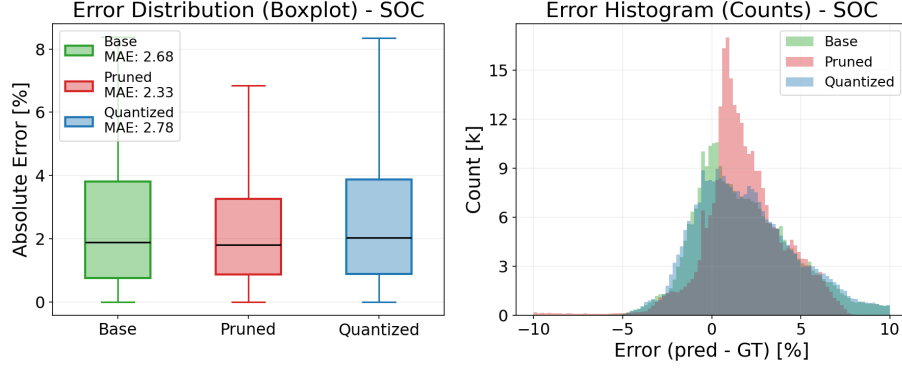


Figure 9: SOC error distributions for Base, Pruned, and Quantized. The left panel compares absolute errors and aggregate MAE. The right panel shows signed errors, $100(\hat{y} - y)$, so that distribution shifts and tails remain visible.

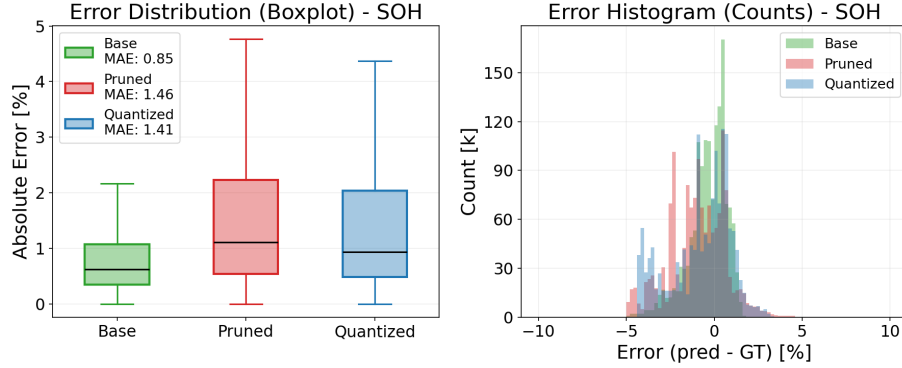


Figure 10: SOH error analysis: MAE distribution (left) and error histogram (right). The three variants retain similar central error ranges, while their tails and aggregate MAE values differ.

8.2. Temporal stability and filter interaction

The ten-segment analysis in Appendix A.2 retains the natural temporal order and does not reset recurrent states between segments. SOC target error rises toward the later part of the sequence for every variant: for example, Base MAE increases from 1.116 to 4.538 percentage points and Quantized MAE from 1.234 to 4.600 percentage points between the first and final tenths. However, the Quantized-to-Base mean absolute deviation changes from 0.332 to 0.271

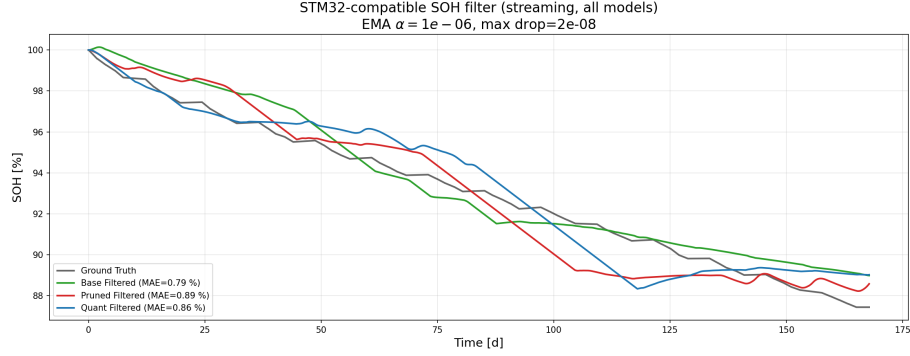


Figure 11: Streaming SOH dashboard with the complete two-stage causal post-processing chain from Section 3.5. The top panel shows the reference SOH and the outputs of the Base (green), Pruned (red), and Quantized (blue) models. The bottom panel depicts the corresponding filtered prediction errors.

percentage points over the same interval and therefore shows no quantization-specific monotonic accumulation. The Pruned-to-Base deviation increases from 0.924 to 2.592 percentage points even though Pruned has the lowest aggregate target MAE. SOH window errors fluctuate with the ageing phase, and neither compressed variant exhibits monotonic model-to-Base divergence across all ten segments. The analysis therefore demonstrates stable relative behaviour throughout the complete evaluated replay and provides a basis for extending validation to further operating sequences.

Appendix A.3 compares unfiltered, stage-1, and final two-stage SOH outputs for the three exported C models on a common input sequence. Stage 1 gives MAE values of 1.677, 1.644, and 1.926 percentage points for Base, Pruned, and Quantized, respectively. Applying stage 2 to the stage-1 output gives final MAE values of 1.476, 1.695, and 1.028 percentage points. The identical input and processing order reveal how filtering changes both the absolute error and the ranking of the model variants. Combining raw and filtered accuracy with the filter response time therefore provides a more complete assessment of the estimator dynamics. The final EMA has an EMA-only 90 % response time of 26.65 days at 1 Hz.

644 8.3. Error Distribution Analysis

645 To quantify the reliability of the estimators, we analyse the statistical dis-
646 tribution of the prediction errors. Figures 9 and 10 present the Mean Absolute
647 Error (MAE) boxplots and the error histograms for both tasks. The SOC error
648 distribution is narrow and approximately centred around zero. The evaluated
649 Pruned model attains the smallest aggregate MAE, while the Quantized model
650 remains close to the Base result with a smaller recurrent-weight footprint. Ap-
651 pendix A.1 shows that the 19 removed SOC units occupy the lowest positions
652 of the defined L2 saliency ranking and compares the central recurrent-weight
653 distributions before and after pruning. This confirms that the implemented cri-
654 terion removes the intended low-saliency channels and identifies Pruned as the
655 most accurate variant for the evaluated model and data partition. Repeated
656 training runs can determine the statistical consistency of this advantage with-
657 out attributing it to a specific regularisation mechanism. For SOH, the error
658 distribution is broader, which is inherent to the difficulty of estimating capacity
659 fade from operational data. However, the histograms show that the bulk of the
660 errors are still contained within a useful tolerance band (typically $\pm 2\%$), making
661 the embedded estimator a viable tool for onboard diagnostics.

662 8.4. Transient Response and Zoomed Analysis

663 Global metrics often hide local failures. To inspect the transient behaviour,
664 we zoom into a high-dynamic pulse discharge phase (Fig. 12) and into a repre-
665 sentative check-up sequence (Fig. 13).

666 In the pulse region (Fig. 12), the models must react to rapid current changes.
667 The quantized model follows the ground truth transients with high fidelity,
668 capturing the voltage recovery phases accurately. The pruned model, having
669 fewer parameters to model the complex electrochemical relaxation, tends to
670 smooth out these sharp features slightly. During the check-up phase (Fig. 13),
671 where the cell undergoes full charge/discharge cycles, all models converge well,
672 demonstrating stability over long integration horizons.

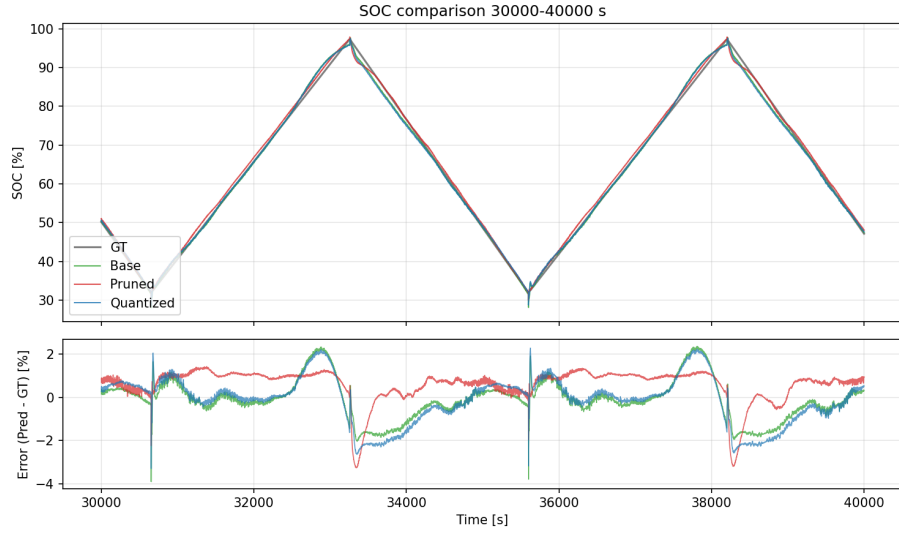


Figure 12: Detailed transient analysis of SOC streaming replay (pulse segment). Response to dynamic load pulses (30k–40k s).

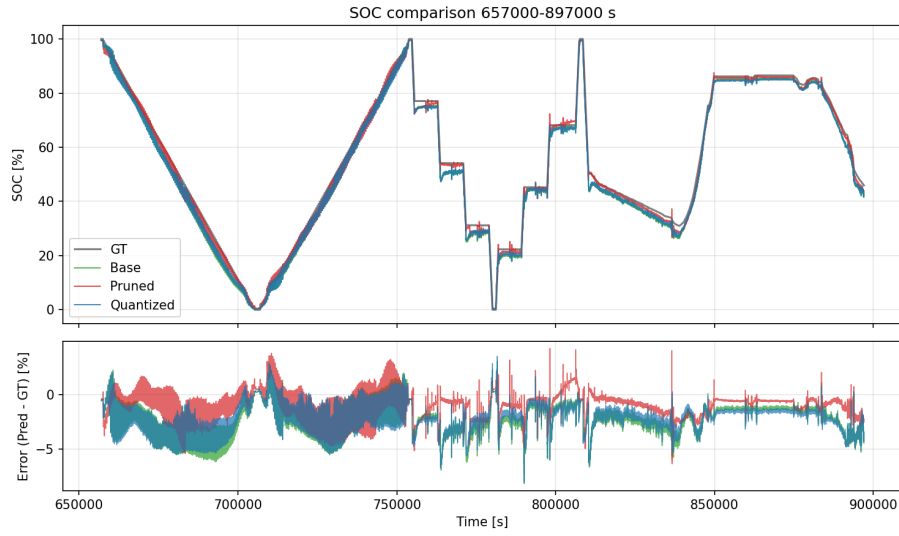


Figure 13: Detailed transient analysis of SOC streaming replay (check-up segment). Behaviour during a capacity check-up (657k–897k s).

673 8.5. Resource Efficiency and Latency

674 Figures 14 and 15 summarise the flash memory, RAM usage, parameter
675 counts, and latency distributions for the SOC and SOH deployments on the
676 STM32H7 microcontroller. For all deployments the model complexity is first
677 characterised on the workstation side by counting parameters in each variant
678 (base, pruned, quantized). Assuming four bytes per weight for floating point rep-
679 resentations and one byte per weight for INT8, this directly yields an estimate of
680 the flash footprint attributable to the neural network weights alone. The second
681 stage of the analysis uses the linker map files of the STM32CubeIDE projects
682 to determine the actual binary size in flash, which also includes code, constant
683 tables, and run-time support libraries. This explains why the measured flash
684 usage in Figure 14 is systematically higher than the idealised "weights only"
685 estimate, and why the gap is larger for the more complex SOH firmware that
686 contains additional diagnostic and filtering logic. The RAM usage shown in the
687 lower right panel of Figure 14 is obtained from on-device measurements of the
688 SOH and SOC firmware and applied analogously to all STM32 deployments.
689 At boot time the firmware reports the size of the statically allocated data and
690 BSS sections, which correspond to persistent variables and network parameters
691 stored in SRAM. During each benchmark inference the minimum value of the
692 stack pointer is tracked, so that the maximum additional stack consumption of
693 the call chain can be recovered. The total RAM requirement reported in the fig-
694 ure is the sum of static data and worst-case stack usage, i.e. an upper bound on
695 the memory needed to execute one inference including the LSTM state updates
696 and UART handling. This procedure captures the fact that pruning reduces
697 both the number of parameters and the recurrent state dimension, whereas the
698 weight-only quantized implementation primarily reduces recurrent-matrix stor-
699 age. Activations, hidden and cell states, biases, and the MLP remain FP32.
700 Figure 14 shows that Quantized produces the smallest flash footprint, while
701 Pruned reduces both the parameter count and recurrent-state dimension. The
702 measured RAM and execution-time results must nevertheless be considered sep-
703 arately because retaining FP32 activations and changing the kernel arithmetic

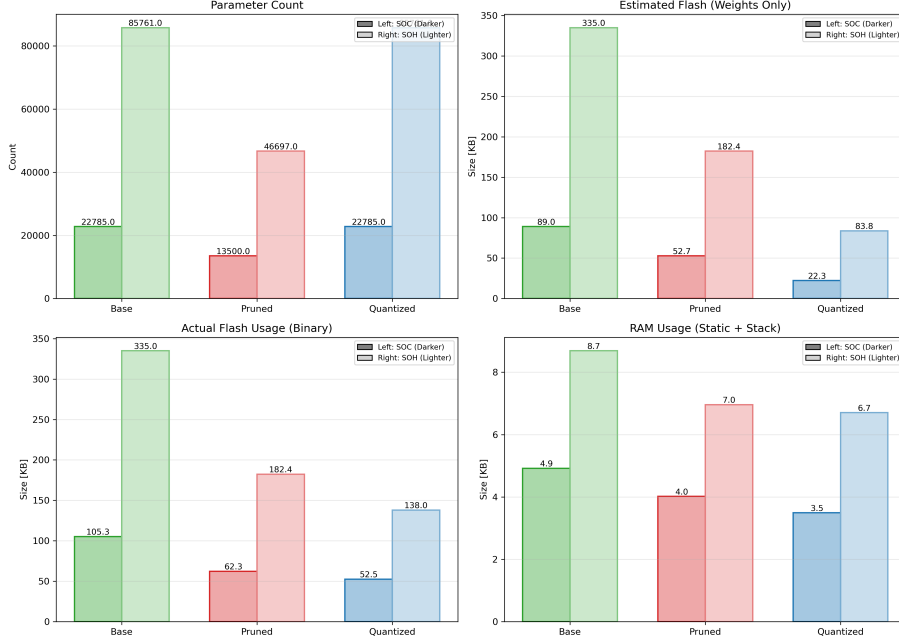


Figure 14: Resource landscape for SOC and SOH models. Comparison of parameter count, estimated Flash, actual Flash, and RAM usage. Dark bars denote SOC, light bars denote SOH.

prevents storage reduction from translating automatically into lower latency. Figure 15 reports the corresponding host-side latency distributions. The latency distributions in Figure 15 reveal a distinct hierarchy. SOC inference requires approximately 0.8 ms to 7 ms, whereas the SOH variants require 12 ms to 30 ms per update. The weight-quantized SOC deployment is slowest because the handwritten mixed-precision kernel applies FP32 row scales during the recurrent matrix products. When UART framing and host processing are included, end-to-end latencies remain below roughly 20 ms for SOC and 50 ms for SOH.

8.6. Composite trade-off summary

To make the multi-objective trade-offs easier to interpret at a glance, we report a composite utility score U that aggregates accuracy and embedded resource indicators into a single, dimensionless index. For task-specific non-

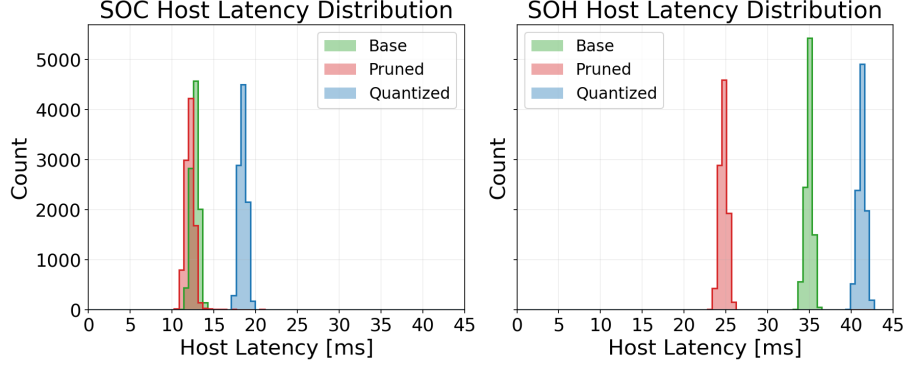


Figure 15: Latency distributions for SOC and SOH models on the STM32H7. Histograms are computed from repeated on-device measurements.

negative weights $\mathbf{w} = (w_A, w_F, w_R, w_E)$ with $\sum_j w_j = 1$, model variant v is scored as

$$U_v(\mathbf{w}) = w_A \frac{\text{MAE}_v}{\text{MAE}_{\text{Base}}} + w_F \frac{F_v}{F_{\text{Base}}} + w_R \frac{R_v}{R_{\text{Base}}} + w_E \frac{E_{\text{est},v}}{E_{\text{est},\text{Base}}}, \quad (27)$$

where MAE is the streaming mean absolute error, F is measured flash, R is measured total RAM (static plus peak stack), and E_{est} is the timing-derived energy proxy. The main comparison uses equal weights, $w_A = w_F = w_R = w_E = 0.25$. In this case, U is the arithmetic mean of the four normalized objectives. By construction $U_{\text{Base}} = 1$ for every valid weight vector. Lower values are preferred. Because E_{est} is proportional to inference time under the fixed-power assumption, the energy term represents timing priority rather than an independent power measurement.

Sensitivity is evaluated over all 1,771 non-negative combinations on a 0.05 grid that sum to one. Appendix A.4 also varies one highlighted objective from 25 to 85 %, sharing the remaining weight equally among the other three. Pruned is top-ranked for 94.64 % of SOC combinations and 53.36 % of SOH combinations. Quantized wins 5.36 % and 17.79 %, respectively, while Base wins 0 % for SOC and 28.85 % for SOH. The ranking is therefore robust for SOC Pruned but application-dependent for SOH: Base becomes preferable when accuracy dominates, whereas Quantized becomes preferable when flash receives high weight.

Table 3: At-a-glance effect of pruning and quantization relative to the corresponding Base model. Values are reported as SOC/SOH. ΔMAE is reported in percentage points (pp) of full scale, where a negative value indicates improved accuracy. Flash, RAM, and estimated-energy changes are relative deltas versus Base. The composite score U uses equal weights in (27) (lower is better).

Variant	ΔMAE [pp]	ΔFlash [%]	ΔRAM [%]	ΔE_{est} [%]	U
Pruned	-0.35/+0.61	-41/-46	-18/-20	-43/-44	0.71/0.91
Quantized	+0.10/+0.56	-50/-59	-20/-23	+399/+29	1.83/1.03

Table 3 summarises the resulting changes for pruning and quantization. Values are reported as SOC/SOH.

8.7. Limited software fault sensitivity

In the missing-update analysis, holding the preceding output for up to 10 % randomly omitted reports changes aggregate MAE by less than 0.001 percentage points. Ten gaps of 3600 samples increase SOC MAE by approximately 0.05 percentage points and have negligible influence on the slowly varying SOH output.

Figure A.20 shows the recurrent response to the one-sample input-buffer faults. Across the SOC variants, median peak deviation ranges from 3.29 to 3.82 percentage points and P95 ranges from 16.18 to 23.81 percentage points. All 30 events per SOC model meet the recovery criterion within 60 s, with median recovery times of 10.0–12.5 s. Quantized follows Base closely, whereas Pruned has a lower median peak but the largest P95 peak. For SOH, median peak deviation ranges from 0.83 to 1.59 percentage points and P95 from 3.37 to 5.85 percentage points. The share recovered within 60 s is 93.3 % for Base, 90.0 % for Pruned, and 86.7 % for Quantized. Among the recovered SOH events, median recovery is 9, 7, and 5 s, respectively.

The target-based measure in Eq. 23 complements these disturbed-clean deviations. For SOC, median ΔMAE_{61} ranges from 0.046 to 0.079 percentage points and P95 from 1.42 to 1.59 percentage points. For SOH, median changes lie between -0.0009 and 0.0012 percentage points and P95 between 0.053 and 0.122

percentage points. The small negative SOH values occur when a disturbance moves an already imperfect prediction closer to the target and are not evidence of improved fault robustness. Table A.4 gives the model-specific values. The test therefore identifies different transient response tails across the compressed variants without indicating persistent SOC divergence or non-finite outputs. Its restricted fault location and event count do not support a hardware-safety claim.

9. Discussion

The experiments show that pruning and quantization affect the SOC and SOH estimators differently. For SOC, the evaluated Pruned model obtains a 0.35-percentage-point lower MAE than Base, while Quantized remains 0.11 percentage points above Base. Together with the L2 saliency and weight-distribution diagnostics, this identifies structured pruning as the more balanced transformation for the evaluated model and split. The experimental design uses one training run and a predefined test split for each task because the focus is the embedded comparison of compression routes. Repeated-seed validation, cross-validation, and matched unpruned fine-tuning can extend the statistical assessment of the observed advantage. The result is therefore reported without assigning it to a specific regularisation mechanism.

For SOH the picture is more nuanced. Targets evolve slowly and are anchored by periodic capacity check-ups, while the displayed online estimates undergo two strong filtering stages. The filter-stage comparison shows that post-processing can alter not only absolute MAE but also the ranking between Base, Pruned, and Quantized. Compression quality should therefore be assessed on both raw and filtered outputs, and filter selection should consider response delay rather than MAE alone.

From an engineering perspective the three deployment variants represent distinct operating points. Base FP32 offers the simplest tool flow but requires the largest flash footprint. Pruning reduces the recurrent state dimension and the corresponding MLP input width, shrinking SOC flash from 105.32 to 62.27 kB

785 and inference time from 1.40 to 0.80 ms. For SOH the corresponding changes
786 are 335.00 to 182.41 kB and 22.73 to 12.72 ms. Within this unchanged FP32
787 kernel family, static MAC ratios and measured time ratios agree closely. SOC
788 Pruned retains 54.9 % of Base MACs and 57.2 % of its time, while SOH retains
789 54.3 % and 56.0 %, respectively.

790 Weight-only quantization gives the smallest flash footprints, 52.48 kB for
791 SOC and 138.00 kB for SOH, but changes the operation types without providing
792 a fully integer execution path. Base and Quantized retain the same 22,080 SOC
793 and 85,120 SOH model MACs. In the current C expressions, each recurrent
794 INT8 weight is cast to FP32 and multiplied by its row scale inside the innermost
795 accumulation, adding 17,920 source-level FP32 scale multiplications for SOC
796 and 68,608 for SOH. Treating one MAC and one additional multiplication as
797 one operation each gives a static Quantized/Base ratio of approximately 1.81
798 for both tasks, whereas the measured time ratios are 4.99 for SOC and 1.29 for
799 SOH.

800 The discrepancy is expected because the count omits conversion, indexing,
801 loop, load/store, nonlinear-function, and cache/Flash effects and because the
802 -00 kernel uses no optimised integer dot-product path. For the larger SOH
803 model, reduced recurrent-weight traffic may offset more of this arithmetic over-
804 head than for SOC. Memory bandwidth was not measured, so this remains
805 an implementation-based interpretation rather than a causal attribution. Ap-
806 pendix A.9 places the static counts beside the measured DWT times.

807 The present kernel prioritises a transparent implementation of the mixed-
808 precision path. Potential improvements include accumulating each row before
809 applying its shared scale, fusing scale and bias work, using DSP-optimised or
810 integer dot-product kernels, and quantizing activations and recurrent states to
811 enable a complete integer path. Operation-level cycle attribution and memory-
812 bandwidth profiling are reserved for future implementation studies. The present
813 results therefore establish total kernel timing for the tested firmware but not
814 the fraction caused by conversion, arithmetic, or memory access.

815 The distinction between system latency, inference time, and estimated en-

ergy is particularly relevant for resource-constrained BMS. The UART-driven tests report end-to-end latency, while on-device inference spans 0.80–6.99 ms for SOC and 12.72–29.21 ms for SOH. The reported E_{est} values are obtained by multiplying these times by an assumed constant 0.5 W. They do not isolate CPU, flash, SRAM, UART, or regulator contributions. Consequently, the results establish timing and memory trade-offs on the STM32H753ZI, but not a component-level energy budget or a conclusion that flash access dominates consumption.

The utility score makes the competing effects of compression directly comparable by expressing accuracy, flash, RAM, and the timing-derived energy proxy relative to the same Base model. A value below one indicates that the combined advantages outweigh the disadvantages for the selected priorities, while a value above one indicates an overall deterioration. With equal weights, $U = 0.71$ identifies SOC Pruned as a balanced improvement, whereas $U = 1.83$ shows that the flash reduction of SOC Quantized does not compensate for its timing overhead. The SOH scores remain close to one, indicating a genuine trade-off without one consistently dominant variant. The weight-grid analysis adds the important information of how stable these conclusions are when application priorities change. SOC Pruned remains preferred across most weight combinations, while the preferred SOH model depends more strongly on whether accuracy or flash receives greater emphasis. The utility analysis therefore supports a transparent selection of the model variant and distinguishes robust recommendations from priority-dependent decisions.

The implementation-level storage reduction follows directly from the architecture. Removing the same number of hidden channels or storing the same recurrent matrices as INT8 therefore produces predictable parameter-count changes for a fixed model topology. The preservation of estimation accuracy depends more strongly on the cell chemistry, operating conditions, reference construction, and training data. This study quantifies the trade-off for JGC LFP cells at 25 °C. Existing embedded studies across different cell chemistries demonstrate that compact data-driven estimators are broadly fea-

sible [11, 24, 49]. Validation with additional cell chemistries is needed to determine how closely their attainable MAE and preferred compression level follow the trends observed here.

The present study examines two post-training compression routes. The first is one-shot structured pruning with short fine-tuning, and the second is recurrent-weight quantization of the original FP32 model. Quantization-aware training and pruning-aware or iterative training can in principle recover accuracy by adapting parameters to the imposed representation [47, 50]. Evaluating these strategies requires a separate training study and is reserved for future work. The current comparison therefore addresses deployable post-training transformations, not the best achievable result of jointly trained compression.

The analytical MAC relation provides a platform-independent estimate of how structured pruning changes model complexity. The measured flash and timing ratios complement this estimate for the evaluated STM32 implementation and build configuration. Their transfer to other embedded platforms depends on clock rate, floating-point and signal-processing support, compiler optimisation, cache, and memory organisation. Benchmarks on additional controller architectures can therefore quantify how these implementation characteristics translate the same model reduction into practical resource savings.

Preparing the derivative features on the host separates feature construction from neural-network inference and allows the measured timing differences to be attributed directly to pruning and quantization. The long-horizon analysis evaluates continuous inference trajectories, while the transient input-buffer analysis perturbs a measurement before inference and follows its effect through the recurrent state. The latter distinguishes immediate peak deviation, recovery, and target-MAE change. It indicates similar SOC behaviour for Base and Quantized, a larger upper-tail SOC response for Pruned, and a larger share of SOH events without confirmed 60-s recovery for Quantized. A complete embedded front-end and safety assessment can build on these results by additionally covering variable sampling, derivative filtering, recurrent-state and weight memory, activations, communication, and MCU memory.

878 10. Conclusion and Outlook

879 This study set out to quantify how two transparent compression routes
880 change the accuracy and embedded cost of stateful SOC and SOH estimators un-
881 der a common STM32 benchmark. Its contribution is an auditable comparison
882 from trained LSTM + MLP models to continuous C inference with consistent ac-
883 curacy, memory, and timing indicators. Estimated energy per inference remains
884 a constant-power timing proxy and is not an independent power measurement.

885 For SOC, Pruned reduces flash by 40.9 %, RAM by approximately 18 %, and
886 inference time by 42.9 %, while its MAE is 0.35 percentage points lower than
887 Base for the evaluated model and split. Quantized reduces flash by 50.2 % and
888 RAM by approximately 20 %, but increases inference time by 399 % and MAE
889 by 0.11 percentage points in the mixed-precision software kernel. For SOH,
890 Pruned reduces flash by 45.5 %, RAM by approximately 20 %, and inference
891 time by 44.0 %, with a 0.61-percentage-point MAE increase. Quantized reduces
892 flash by 58.8 % and RAM by approximately 23 %, while increasing inference
893 time by 28.5 % and MAE by 0.56 percentage points. Under the fixed-power
894 assumption, the energy proxy changes by the same percentages as inference
895 time.

896 These results identify structured pruning as the more balanced operating
897 point for the tested firmware, whereas recurrent-weight quantization is attrac-
898 tive primarily when flash capacity dominates and its dequantization overhead
899 is acceptable. Ten consecutive replay segments show no quantization-specific
900 monotonic deviation from Base. After transient input-buffer bit flips, every
901 SOC variant recovers in all 30 events within 60 s, while 6.7–13.3 % of the SOH
902 events do not meet the defined recovery criterion within this horizon. The
903 SOH filter analysis further shows that compression must be assessed together
904 with post-processing delay. The evaluated LFP trajectories, model configura-
905 tions, STM32H753ZI implementation, and compiler settings establish the trade-
906 off under one complete deployment scenario. Validation across additional cell
907 chemistries, embedded platforms, model initialisations, and internal fault classes

908 can determine how broadly the observed trends transfer.

909 The first priority for future work is to optimise and profile the mixed-
910 precision kernel through row-scale hoisting, operator fusion, DSP-oriented dot
911 products, and activation/state quantization, while measuring per-model board
912 power with external instrumentation. A second priority is external validation
913 across random seeds, temperatures, cell chemistries, and embedded platforms,
914 including direct comparison with quantization-aware and pruning-aware train-
915 ing. Finally, internal weight, activation, recurrent-state, sensor, communication,
916 and MCU fault injection is required before safety-related claims can be made.
917 Joint pruning and quantization remains a promising route for combining a re-
918 duced state dimension with compact recurrent storage.

919 **Appendix A. Supplementary compression and robustness analyses**

920 This appendix presents supporting analyses of pruning structure, tempo-
921 ral stability, filter interaction, utility sensitivity, transient input faults, model-
922 complexity scaling, mixed precision, and kernel operation counts.

923 *Appendix A.1. Structured-pruning diagnostics*

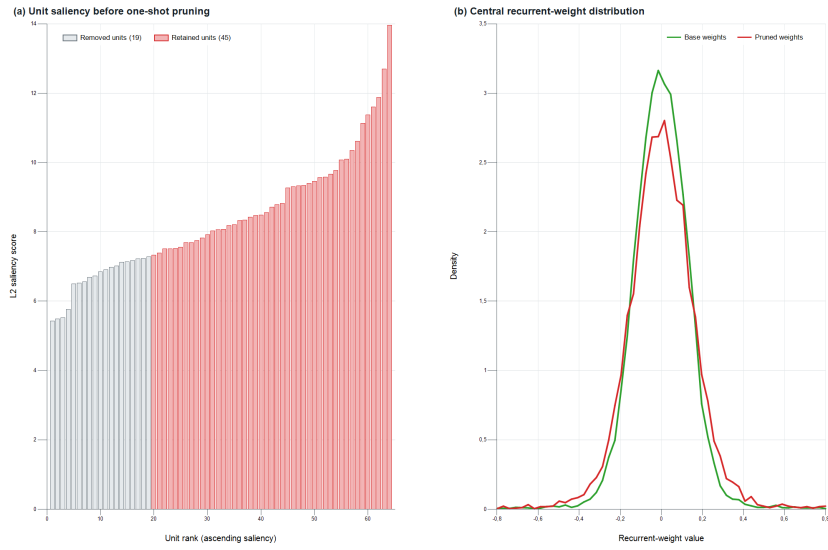


Figure A.16: Diagnostics for the one-shot SOC pruning operation. (a) L2 saliency from Eq. 16, sorted in ascending order. The 19 lowest-ranked units are removed and 45 units are retained. (b) Density of the central recurrent-weight range for the Base and Pruned models. The analysis confirms application of the specified ranking criterion and makes the resulting weight selection transparent. Repeated training runs can extend this structural evidence toward a statistical assessment of the observed MAE difference.

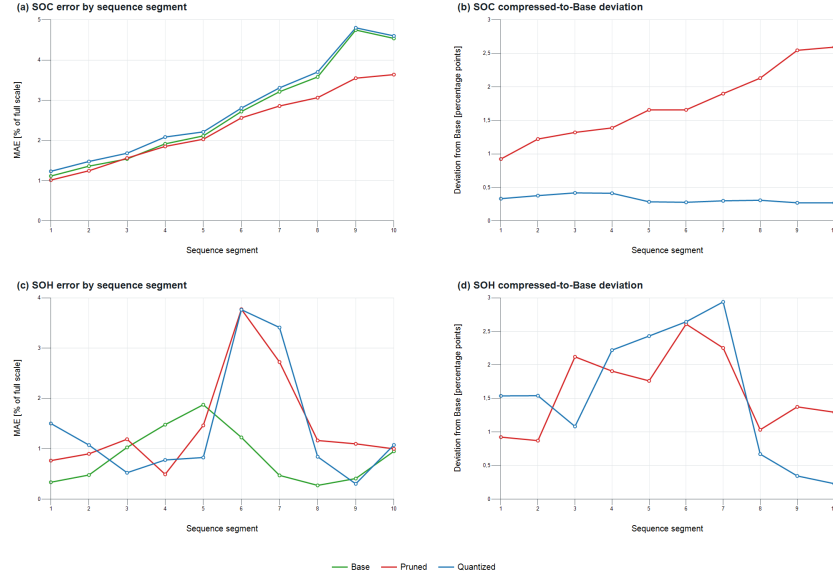


Figure A.17: Temporal error and model-to-Base deviation over ten consecutive, equal-count segments of the full-stream replay. Recurrent states are not reset at segment boundaries. Panels (a) and (c) show target MAE for SOC and SOH. Panels (b) and (d) show the mean absolute deviation of Pruned and Quantized from Base. The absence of monotonic late-sequence divergence demonstrates stable relative behaviour across the complete evaluated replay. Further sequences can extend the tested operating horizon.

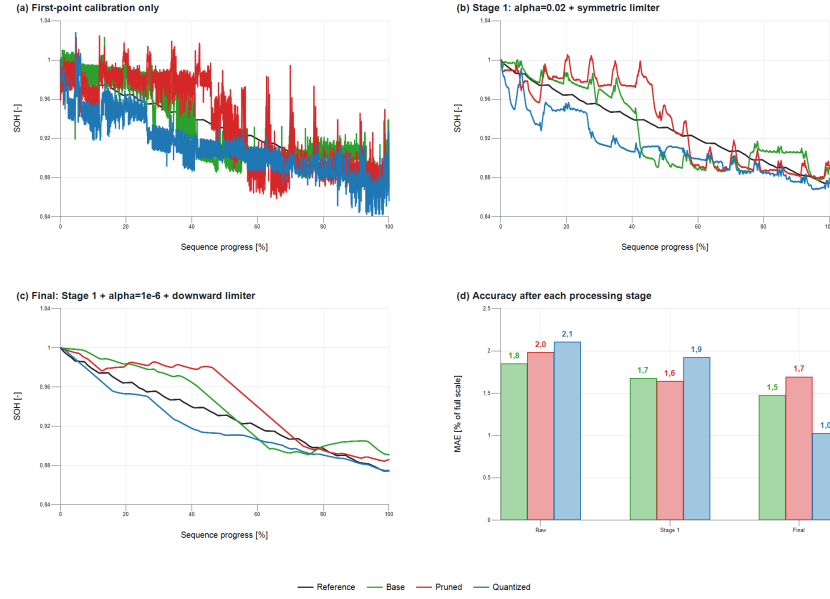


Figure A.18: Comparison of SOH compression and filtering for the Base, Pruned, and Quantized C models on a common input sequence. (a) Outputs after first-point alignment but before temporal filtering. (b) Stage 1 with $\alpha_1 = 0.02$ and the symmetric limiter. (c) Final sequential output after applying stage 2 with $\alpha_2 = 10^{-6}$ and the downward limiter to stage 1. (d) MAE after each processing stage. At 1 Hz, the EMA-only 90 % response times are approximately 114 s and 26.65 days, respectively. The nonlinear limiters have no single cutoff frequency. All three variants use the same input and processing order to isolate the interaction between compression and filtering.



Figure A.19: Sensitivity of the utility ranking to application priorities. Panels (a) and (b) vary the highlighted objective from 25 to 85 % and divide the remaining weight equally among the other three objectives. Each square identifies the lowest-utility model. At 25 %, all four objectives have equal weight. Panel (c) gives the winning share across all 1,771 non-negative 0.05-spaced weight combinations that sum to one. The energy objective uses the timing-derived E_{est} proxy and is therefore not an independent power measurement.

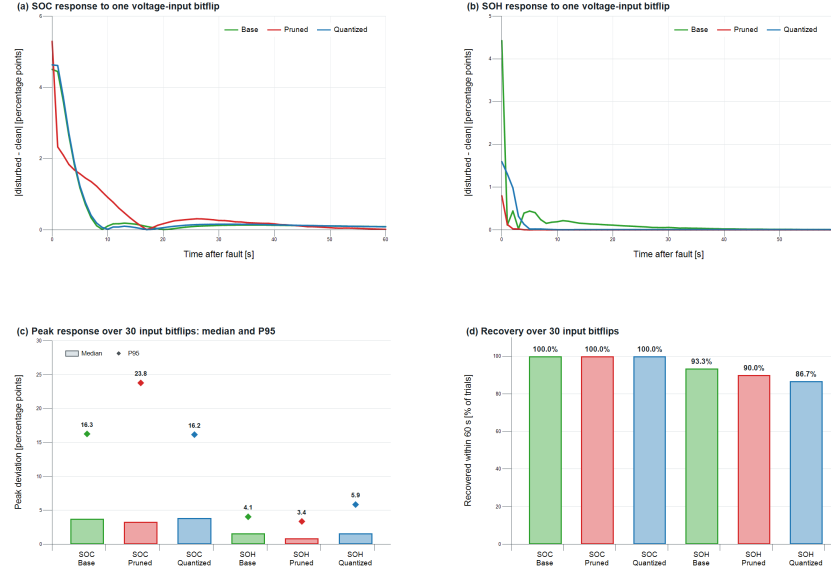


Figure A.20: Stateful response to transient software input-buffer corruption. Thirty independent events per task and model flip the most significant FP32 mantissa bit in voltage, current, or temperature for one 1-Hz sample, with ten events per input signal. The clean and disturbed branches start from the same recurrent state and then receive 60 identical clean samples without a reset. (a) and (b) show the absolute disturbed-clean deviation for one representative voltage event. (c) summarises the median peak and P95 peak across all 30 events. (d) gives the fraction that meets the sustained recovery criterion within 60 s. The figure characterises propagation through the recurrent state, while Table A.4 reports the corresponding change in target MAE.

Table A.4: Change in target MAE over the 61-sample fault and recovery window. Values are percentage points and summarise 30 independent input-bitflip events per task and model. The change is defined by Eq. 23. Negative values indicate that the disturbed output happens to lie closer to the target in part of the window.

Task	Model	Median ΔMAE_{61} [pp]	P95 ΔMAE_{61} [pp]
SOC	Base	0.0740	1.4235
SOC	Pruned	0.0464	1.5867
SOC	Quantized	0.0785	1.4314
SOH	Base	-0.0009	0.1224
SOH	Pruned	-0.00004	0.0533
SOH	Quantized	0.0012	0.1055

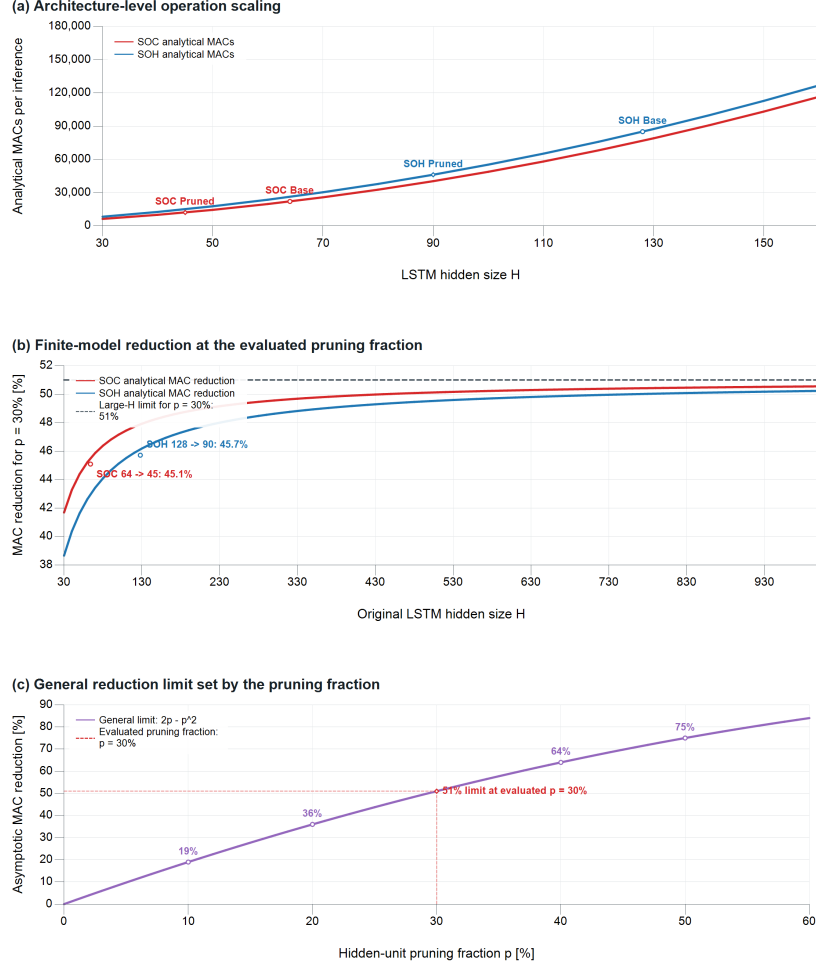


Figure A.21: Architecture-level operation scaling from Eq. 14. (a) Analytical MACs per streaming inference over LSTM hidden size, with markers for the implemented SOC and SOH Base and Pruned dimensions. (b) Finite-model reduction for the evaluated pruning fraction $p = 0.30$, including the implemented SOC and SOH points and the 51 % large- H limit. (c) General relation between the hidden-unit pruning fraction and the asymptotic reduction $2p - p^2$. The highlighted 30 % operating point has a theoretical limit of 51 %, while the corresponding limits for 10, 20, 40, and 50 % pruning are 19, 36, 64, and 75 %. Linear input and MLP terms together with the fixed output term keep finite models below the respective limit.

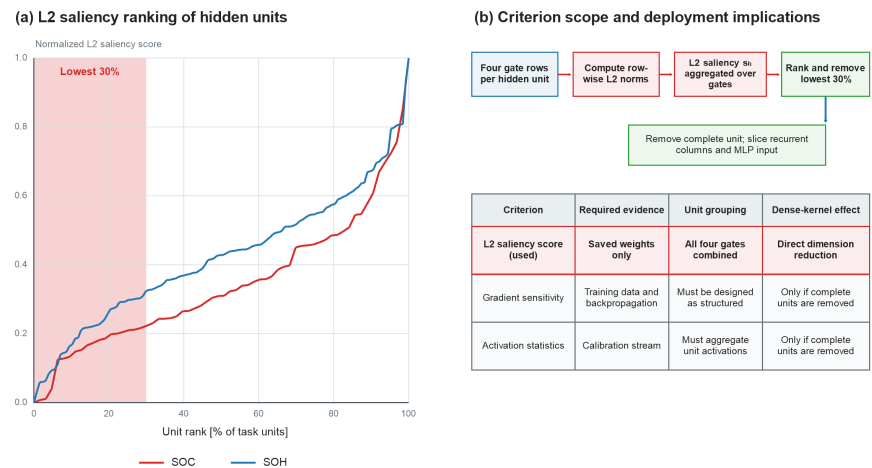


Figure A.22: Scope of the structured pruning criterion. (a) Normalised L2 saliency ranks for the SOC and SOH Base hidden units. The lowest 30 % define the task-specific removal sets. (b) The implemented path aggregates row-wise L2 norms across all four gates and removes complete recurrent units together with associated recurrent columns and MLP inputs. The comparison makes the reproducibility and dense-kernel benefit of the L2 criterion explicit and provides a reference for future comparisons with gradient- and activation-based rankings.

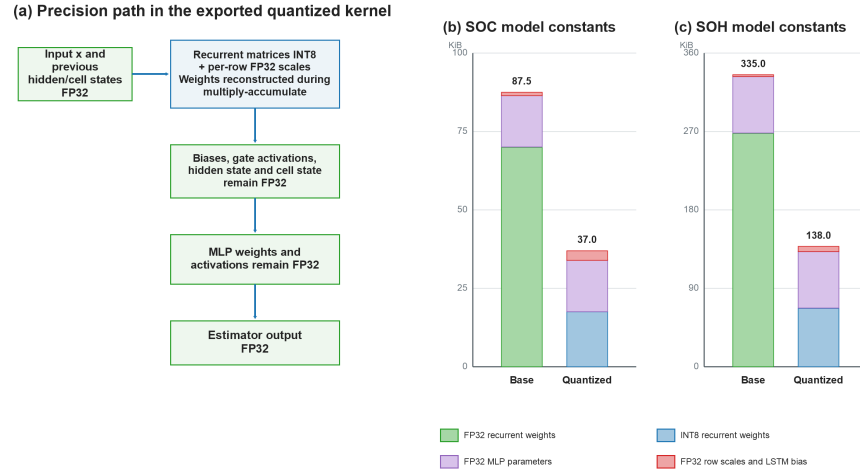


Figure A.23: Verified precision path and raw model-constant composition. (a) Only recurrent input-to-hidden and hidden-to-hidden matrices are stored as INT8. Per-row scales, biases, gate activations, hidden/cell states, the MLP, and the estimator output remain FP32. (b,c) Analytical model constants for SOC and SOH Base and Quantized variants. The comparison shows how concentrating quantization on the dominant recurrent matrices reduces persistent model storage and where activation and state quantization could provide additional runtime savings.

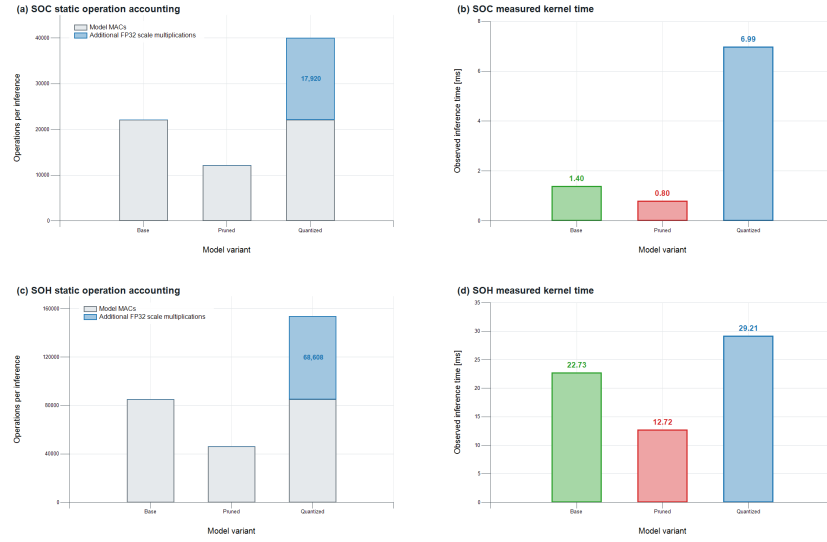


Figure A.24: Static source-level operation accounting beside measured STM32H753ZI kernel time. Panels (a,c) show model MACs and the additional FP32 row-scale multiplications implied by the current per-weight C expressions. Panels (b,d) show DWT measurements averaged over 10,000 streaming inferences. Their comparison reveals where analytical operation counts explain the timing trend and where conversion, indexing, loop, load/store, nonlinear-function, compiler, and memory-system effects create additional optimisation opportunities.

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946 **CRedit authorship contribution statement**

947 Florian Rzepka: Conceptualization, methodology, software, validation, for-
948 mal analysis, investigation, data curation, writing–original draft preparation,
949 writing–review and editing, visualization.

950 Julia Kowal: Resources, supervision, project administration, funding acqui-
951 sition, writing–review and editing.

952 **Declaration of competing interest**

953 The authors declare that they have no known competing financial interests or
954 personal relationships that could have appeared to influence the work reported
955 in this paper.

956 **Declaration of generative AI and AI-assisted technologies in the manuscript**
957 **preparation process**

958 During the preparation of this work the author(s) used ChatGPT, DeepL,
959 and Grammarly in order to improve the linguistic quality and style of the
960 manuscript. After using these tools, the author(s) reviewed and edited the
961 content as needed and take(s) full responsibility for the content of the published
962 article.

963 **Data availability**

964 The LFP DoE Cycle Ageing Dataset used in this work can be obtained from
965 the DOI repository in [15]. The landing page provides licensing information and
966 download instructions.

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